Following to Lead: Dominant Followership in Sequential Price Competition

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Abstract

A dominant platform might be expected to use its market position to charge more than rivals, even for the same products. Amazon does not. Using synchronized high-frequency cross-platform pricing data, I find that Amazon is usually priced at or below the lowest rival price. Its distinctive role appears instead in how it responds to rival price changes, where I document three patterns. First, after a rival raises its price, Amazon's probability of raising its own price more than doubles, whereas rivals show no detectable response when Amazon moves first. Second, when Amazon responds, it matches the rival's new price. Third, matched rival price increases remain in place longer than unmatched ones, while matched decreases show no comparable persistence pattern. I call this strategy dominant followership. A sequential Bertrand model modified to break ties in favor of the dominant firm shows why dominant followership can raise prices above the simultaneous-move Nash benchmark. Because the dominant firm keeps most demand at price parity, it may prefer matching a rival's higher price to undercutting it. The rival is willing to raise price first when that match is predictable; once the price is matched, it can persist when the rival's ability to respond again is restricted. Using simulation studies, I show that Q-learning algorithms discover dominant followership endogenously: the rival's algorithm learns to initiate higher prices, nearly tripling the market price, yet the dominant firm captures 91 percent of the resulting profit gains. The alignment of demand and response advantages is crucial: reassigning the reliable-responder role to the rival can reverse the price-raising effect. This speaks to the FTC's Project Nessie allegation: individually optimal responses can teach rival algorithms that raising price is profitable, even without explicit coordination. The main insight is that market power need not operate through price leadership: it can operate through anticipated response. Modern algorithmic price competition among asymmetric firms should therefore be modeled as a sequential response process, in which who responds matters as much as the price itself.

Keywords: **Algorithmic Pricing, Artificial Intelligence, Sequential Price Competition, Demand Asymmetry, Reinforcement Learning**

JEL Classification: **L13, L41, C63, C73, L81**

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1 Introduction

A firm with a demand advantage, one that can retain customers even when it does not offer the lowest price, is ordinarily expected to translate that advantage into a price premium. The online retail data I study run against this expectation: despite its dominant market position, Amazon is usually priced at or below the lowest rival price for the same product.

Does this reflect strong competition, or something else? Price levels alone cannot answer this question. In markets where prices are visible, firms do not choose prices only once or only simultaneously; they observe rivals' prices and respond over time. A dominant firm's market position may therefore matter not only through the price it posts at a point in time, but also through how its responses affect the profitability and durability of rival price changes.

This response channel is becoming more important as pricing itself becomes automated. Pricing algorithms allow firms to automate not only the prices they charge, but also how they respond when rivals adjust prices. These algorithmic price-response functions are becoming increasingly important as AI-driven pricing tools make responses more scalable and adaptive.[1] They have already attracted regulatory attention: the U.S. Federal Trade Commission (FTC) has accused Amazon's internal pricing algorithm "Project Nessie" of guiding the entire market toward higher prices through algorithmic price responses (Federal Trade Commission 2024).[2]

The concern extends beyond Amazon and any single enforcement action. Once responses are systematic, they become part of the environment from which rival algorithms learn, especially when the responder is a dominant firm whose price serves as a salient benchmark. If rivals learn to anticipate that response, dominance need not operate through leadership: a dominant firm can shape market prices by becoming a predictable follower. I ask whether this response channel can raise market prices even while the dominant firm's own price levels appear competitive.

Understanding when this response channel matters requires moving away from the symmetric-firm structure common to standard game-theoretic models of simultaneous price competition. I

[1] Among 419 B2B pricing executives and decision-makers, 65 to 85 percent reported that their organizations expected to adopt generative and agentic AI in pricing within one to three years; see McKinsey, "B2B pricing: Navigating the next phase of the AI revolution," April 7, 2026.

[2] A related California enforcement action alleges coordinated price increases in which one retailer raised its price and another subsequently matched the higher price. Unlike the mechanism studied here, which operates through independently chosen price responses and does not require coordination, the California filing alleges vendor-mediated agreements among Amazon and competing retailers. See *People v. Amazon.com, Inc.*, Memorandum in Support of Motion for Preliminary Injunction (Cal. Super. Ct. 2026), pp. 2–3.

introduce two asymmetries. The first is demand advantage: one firm retains a larger share of demand when prices are equal. Amazon is a natural example, allegedly accounting for more than 82 percent of gross merchandise value in the online-superstore market defined in the FTC complaint. The second is response asymmetry: one firm can respond to rivals' price changes more systematically and reliably than another. These asymmetries need not favor the same firm. However, the price-raising channel works when they make the demand-advantaged firm the follower. If the follower role shifts to the other firm, the channel can break down.

Using a custom scraper that tracks Amazon's first-party prices against every rival platform selling the exact same product, every two hours over eight months, I document how Amazon and its rivals respond to each other's price changes in more than one million synchronized price observations. Amazon's response unfolds in three parts, together showing when rival-initiated higher prices are more likely to persist. First, it reacts far more often than rivals do: after a rival-initiated price increase, Amazon is 12.9 percentage points more likely to raise its own price within 48 hours, 3.5 times the response of other major platforms, who respond by only 3.7 points. When Amazon itself moves first, other major platforms show no detectable response. Second, when Amazon does react to a rival increase, it typically matches rather than undercutting the rival's new price: 40.3 percent of its responses land exactly at that price. Third, when Amazon matches a rival's higher price, those rival increases are more likely to persist: matched rival price increases are 15.8 percentage points more likely to remain in place after two weeks than unmatched increases, while matched decreases show no such effect. I call this strategy "dominant followership." When rivals can anticipate that the dominant firm will match a higher price, raising price first can become profitable for the rival.

The dominant followership documented in the data can be rationalized in a simple Bertrand model with one seemingly minor modification: one firm has a demand advantage at price parity. One way to microfound this assumption is a lexicographic decision rule: consumers choose the lower price, but when prices are equal, they favor the dominant platform. This keeps the model close to homogeneous-products Bertrand competition, while allowing platform advantages to matter. The key insight is that this advantage gives the dominant firm little incentive to undercut the rival's higher price; it may instead prefer to match it, monetizing the demand it already holds. The rival, holding only a minority of that same tied-price demand, has every incentive to undercut instead.

But this undercutting incentive pushes prices toward marginal cost only when the rival can react quickly. A market friction that limits the rival's ability to react can therefore sustain equilibrium prices above the simultaneous-move Nash equilibrium. The single-period model captures this friction by allowing only one response before the market clears. The dynamic model then constructs a Markov-perfect equilibrium in which dominant followership persists: the rival raises price first, the dominant firm observes and matches, and the market remains at the higher matched price. This sequence is sustained by two different response frictions: the dominant firm must have frequent enough response opportunities to make rival initiation profitable, while the rival must have limited enough later response opportunities to deter both dominant-firm initiation and rival undercutting.

The theoretical model explains why dominant followership can soften competition when firms understand the incentives they face. In real-world online retail, however, retailers set prices across large assortments where demand conditions and rival pricing patterns differ across products and evolve over time. The learning question is therefore whether pricing algorithms in asymmetric market roles can learn policies that generate this response pattern without being programmed to do so. I use Q-learning simulations in which firms update pricing policies from realized profit feedback, but are not programmed with the dominant-followership strategy. The simulations show that the mechanism can emerge endogenously. Compared with a baseline in which neither demand advantage nor response asymmetry is present, demand asymmetry alone raises the learned market price by only about 12 percent, asymmetric response friction alone by 37 percent, and both combined by nearly 70 percent. The alignment of the two asymmetries matters: assigning the low-friction responder role to the rival instead breaks the mechanism, pushing the learned price below either single-asymmetry case.

Strikingly, the algorithm that learns to raise prices and the firm that benefits from that learning need not be the same: the rival's algorithm drives the price increase, while the dominant firm captures most of the reward. This is surprising because the theoretical mechanism depends on the dominant firm's predictable match, suggesting that its own algorithm should be the one that has to learn the mechanism. To identify whose learning matters, I let one firm at a time use Q-learning while the other follows a simpler, myopic best-response pricing rule. The decomposition confirms the reversal: when only the dominant firm learns, the market price remains low; when only the rival learns, it reaches 89 percent of the price level achieved when both firms learn. The intuition is a stark asymmetry in what each firm has to figure out. For the dominant firm, matching a higher rival price is a one-shot decision, requiring no experimentation. For the rival, initiating that higher price is a gamble: it only learns whether the bet pays off after repeatedly absorbing the short-run cost of raising price first and waiting to see how the dominant firm responds. Yet once the rival's algorithm learns that initiation can be profitable, the dominant firm captures 91 percent of the resulting profit gains. This has a direct bearing on cases like the Project Nessie allegation: if a rival's own algorithm learns to raise prices, scrutiny focused only on the dominant platform's pricing algorithm may be looking in the wrong place.

The main insight of this paper is that market power need not operate through price leadership. In markets where firms observe and respond to one another's prices, a dominant firm can affect market prices by becoming a predictable follower. This shifts the central object of analysis from the price a firm posts at a point in time to the sequence of price changes: who moves first, who responds, and how demand is allocated once prices are matched. Simultaneous-move models miss this channel because they remove the responder role through which demand advantage shapes response incentives and learning. They therefore fail to approximate long-run outcomes when firms differ in both demand advantage and response friction.

The paper connects three literatures through one mechanism: dominant followership in sequential price competition among asymmetric firms. First, I provide new evidence on a response margin that standard pricing data usually miss. Using novel high-frequency cross-platform data, I show that Amazon plays a distinctive responder role: rival price increases are more durable when Amazon matches them. Second, I introduce dominant followership as a mechanism through which demand asymmetry can soften competition without dominant-firm leadership. If rivals expect the demand-advantaged firm to match reliably enough, they may be willing to raise price first. Third, I show that this mechanism can emerge endogenously from algorithmic learning. The learning result separates initiation from incidence: the rival's algorithm generates the price increase, while most of the gains accrue to the dominant firm.

These findings change how algorithmic pricing should be evaluated. Dominant followership need not look anticompetitive in isolation. The dominant firm is not necessarily leading prices upward, communicating with rivals, or committing to a common pricing rule. It may simply be responding to a rival's price as an individually optimal response. The concern is that, once these responses become systematic, they can enter rival algorithms' learning environments and raise market-wide prices without communication or agreement. The resulting welfare concern is that individually optimal responses can shift surplus from consumers toward firms through a channel that standard price-level screens may miss. Regulators and modelers should therefore pay attention not only to price levels, but also to response patterns, timing, and which firms' algorithms learn from those responses. For a firm weighing whether to adopt a learning algorithm, the results carry a specific caution: it may be your own algorithm that learns to raise prices, while a dominant rival is the one who profits from it.

Related Literature. Empirically, the paper shifts attention in the algorithmic-pricing literature from price levels to price responses. Existing empirical work primarily studies price levels, margins, and whether algorithm adoption changes pricing outcomes in particular markets, including automobiles, gasoline, housing, online marketplaces, and earlier digital-pricing settings (Sudhir 2001; Kannan and Kopalle 2001; Assad, Clark, Ershov, and Xu 2024; Calder-Wang and Kim 2023; Musolff 2024; Zhao and Berman 2025). Less is known about algorithmic responsiveness: whether firms react to rivals' price changes within a given response window, and how those reactions vary across firms with different degrees of demand advantage. A related e-commerce literature documents price dispersion, Buy Box competition, and algorithmic repricing within online marketplaces, especially Amazon Marketplace and online retail (Cavallo 2018; Chen, Mislove, and Wilson 2016; He, Hollenbeck, and Proserpio 2022; Lam 2023; Hanspach, Sapi, and Wieting 2024; Musolff 2024). The Project Nessie allegations point to a different empirical object: cross-platform price responses, in which one online retailer's price change may affect pricing decisions by other retailers. Studying this margin requires observing identical products across platforms at high frequency, which is difficult because Amazon's prices are relatively visible while rival retailers' prices are harder to observe. I address this gap by constructing synchronized high-frequency pricing data that measure how price changes propagate sequentially across retailers.

Theoretically, the paper builds on models of algorithmic pricing and learning. Canonical and recent studies show that independently learning algorithms can generate supracompetitive outcomes in repeated pricing environments with simultaneous or concurrent price choices (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024;

Wang, Huang, Singh, and Srinivasan 2025). These settings are less suited to posted-price markets, where prices are observable and firms can react to rivals' prices at different times, with demand realized between price adjustments. I shift attention from algorithmic collusion in simultaneous or symmetric pricing environments to algorithmic responsiveness under asynchronous price adjustment among asymmetric firms.

The closest related work relaxes symmetry in algorithmic responsiveness. Brown and MacKay (2025) show that speed and commitment on the supply side can allow one algorithmic firm to induce supracompetitive prices from a slower or myopic rival. The dominant-followership mechanism differs from theirs because response asymmetry matters through its alignment with demand advantage: rivals are willing to raise price first only when they anticipate that the demand-advantaged firm is likely to respond and, conditional on responding, prefers matching to undercutting. The key object is therefore not commitment by the fast firm alone, but the alignment of demand advantage and responder role, which the rival can learn to anticipate.

This focus on response roles also connects the paper to dynamic oligopoly. Maskin and Tirole (1988) provide a canonical model of alternating-move price competition, in which firms condition pricing decisions on rivals' current prices, and Klein (2021) extends this sequential pricing framework to Q-learning agents. My framework builds on this sequential structure but changes the economic role of moving second: when the follower has a demand advantage, moving second can be more valuable than leading because the follower's predictable match makes rival initiation profitable.

The paper also relates to the empirical literature on price leadership. Byrne and de Roos (2019) show that dominant firms can use leadership and price experiments to create focal points and coordinate market prices. In contrast, the empirical patterns in this paper motivate a focus on how dominant firms respond, rather than how they lead.

Dominant followership also differs from two forms of matching studied in earlier work. Classic repeated-game work studies reciprocal strategies such as tit-for-tat, in which cooperation is sustained because firms punish deviations from past behavior (Axelrod and Hamilton 1981). The price-matching-guarantee literature studies explicit policies under which firms commit to match rivals' lower prices (Edlin 1997; Corts 1997; Hviid and Shaffer 1999; Chen, Narasimhan, and Zhang 2001). Matching in my setting is neither reciprocal punishment nor an advertised commitment.

It is an observed algorithmic response pattern whose competitive effect comes from the demand advantage of the firm doing the matching.

Finally, the paper connects algorithmic pricing to the literature on demand-side frictions. Search frictions, product differentiation, loyalty, and switching costs can soften price competition by making consumers less willing or able to switch across firms (Hotelling 1929; Diamond 1971; Varian 1980; Narasimhan 1988; Beggs and Klemperer 1992). This literature mainly studies how such frictions affect price levels and market power. I show that they can also shape the roles firms play in sequential algorithmic competition. In my setting, demand advantage matters not mainly by supporting a price premium, but by changing which responses are profitable and learnable once prices are matched.

2 Sequential Price Responses Across Platforms

This section builds the empirical case for dominant followership. A platform with a demand advantage, meaning consumers favor it for reasons other than price, such as brand trust, membership programs, or convenience, should be able to price above its rivals without losing much business. Amazon is a natural setting to look for this kind of advantage: it is the dominant platform in this market, allegedly accounting for more than 82 percent of gross merchandise value in 2022 under the online-superstore market definition used in the FTC complaint (Federal Trade Commission 2024). Yet Amazon is not typically the high-price platform; if anything, it is usually priced at or below its rivals. This is the empirical puzzle: why would the platform best positioned to leverage a demand advantage instead compete so aggressively on price? I answer this puzzle by turning from Amazon's price level to its price responses: Amazon's demand advantage shows up not in charging more, but in how it reacts when a rival does. I show that Amazon is unusually likely to raise price soon after a rival raises its own, typically by matching the rival's new price rather than undercutting it. Rival increases that Amazon matches are also more likely to persist. This response pattern lets Amazon benefit from higher prices without relying on being the first mover.

2.1 High-Frequency Cross-Platform Pricing Data

Studying price responses requires data that capture a part of competition usually hidden in standard pricing data: the order in which prices change across competing platforms. To recover this order, I build a high-frequency scraping system that tracks identical products across the full set of identified platforms selling each matched variant, collecting prices within synchronized time windows.

Three design features are central. First, I match products at the exact variant level, including SKU, model specifications, and color, so observed price differences reflect platform pricing rather than product heterogeneity. Second, prices for a given product are collected across platforms within the same time window, at roughly two-hour frequency. This synchronization makes it possible to compare prices for the same product across platforms at nearly the same moment. In 91.3% of price-change bins, exactly one eligible platform changes price, so the panel can recover an order of moves rather than only comparing prices at a point in time.[3] Third, for Amazon and Walmart, where the scrape observes marketplace seller identity, I restrict observations to first-party platform listings. This is important because the paper studies platform pricing, not seller pricing within a marketplace. For the other major platforms, the scraped pages do not provide a separate marketplace seller field, so I treat the observed listed price as the platform price.

The panel covers a broad set of consumer electronics and home-appliance categories, including headphones, wearables, smart-home devices, and vacuum and floor-care products, from November 2024 to June 2025.[4] I begin with prominent Amazon-listed products within these categories and keep only products also sold by at least one non-Amazon platform; because the relevant competitors differ across products, the resulting platform set varies and spans both broad-selection online superstores and narrower specialty or brand-specific retailers. After dropping observations with missing prices or marked out of stock, the resulting platform-pricing panel contains 216 matched products across 52 identified platforms, 792 product-platform pairs, and about 1.02 million in-stock price observations. Products are observed on 3.67 platforms on average, and the average scrape interval is 2.3 hours. The product set covers a broad range of price points and demand levels: median product prices range from below \$50 to above \$500, and Amazon's "Bought in Past Month"

[3] I define a price update as a change from the platform's most recent observed price for the same product. To avoid treating long gaps as immediate responses, I use only cases where the prior observation occurs within the preceding one to six hours.

[4] Categories are selected based on sufficient price variation to study adjustment behavior.

demand proxy ranges from 50 to 100,000.

2.2 Empirical Puzzle: Amazon's Price Position

If Amazon's demand advantage translated into pricing power in the usual way, it should rarely be the cheapest option on the page. Figure 1 plots Amazon's price relative to the lowest rival price, using observations in which Amazon and at least one rival list the same product.

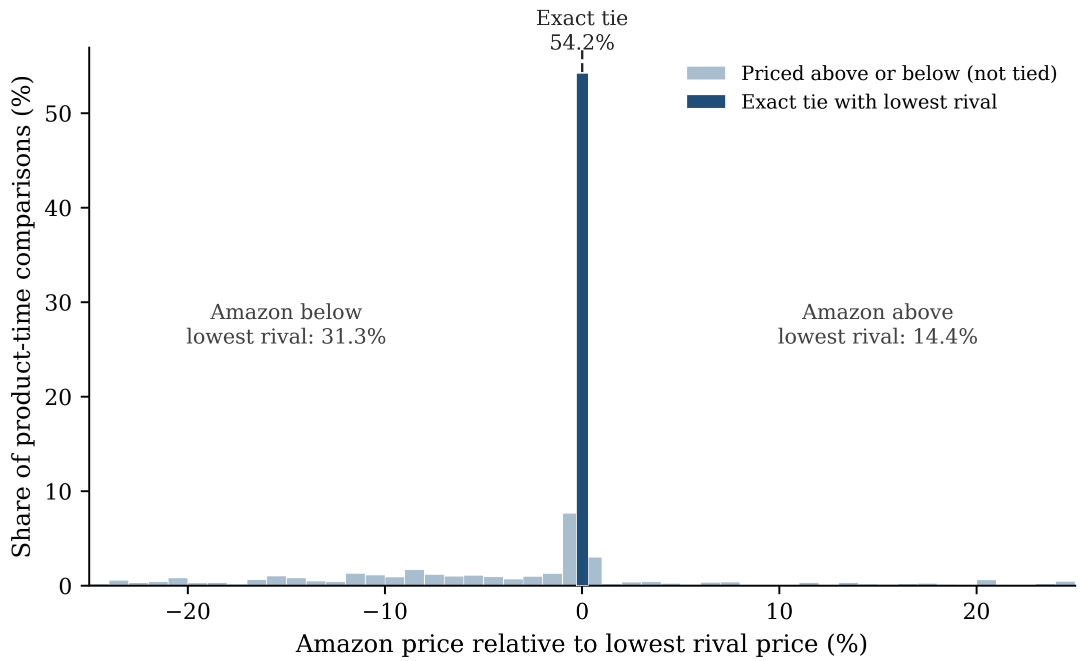

Figure 1. Amazon's Price Relative to the Lowest Rival Price.

Notes: The figure plots Amazon's price relative to the lowest rival price across product-time comparisons in which Amazon and at least one rival have listed prices. A value of zero denotes an exact tie with the lowest rival. Negative values mean Amazon is priced below the lowest rival; positive values mean Amazon is priced above the lowest rival. The figure plots gaps between -25% and 25%; 10.4% of comparisons fall outside this range and are not shown.

Amazon is exactly tied with the lowest rival in 54.2% of comparisons, priced below it in 31.3%, and priced above it in only 14.4%; more than three quarters of comparisons fall within 10% of the lowest rival price. This pattern is stable over time: Amazon's weekly lowest-price share ranges from 77.5% to 88.9% across the sample period, consistently above the weekly mean lowest-price share among major rivals (Walmart, Target, Best Buy, and Home Depot), which ranges from 40.4% to 60.2%.

Amazon's distinctive market role, then, is not expressed through a persistently higher price. It shows up instead in how Amazon responds when rivals move first.

2.3 Amazon Is More Likely to Raise Price After Rival Increases

Amazon is unusually likely to raise price soon after a rival does. Table 1 compares, for each Amazon-product observation, moments when a rival has just raised price against moments when no rival has, using whether Amazon raises price within the next 48 hours as the outcome.[5] Amazon increases within 48 hours in 10.2% of cases when no rival has just raised price, compared with 30.7% of cases when a rival has. The difference is 20.4 percentage points. The same pattern remains after controlling for product, platform, week, and day-of-week differences: Amazon is 15.9 percentage points more likely to raise price after a rival increase ($p < .001$; Panel A of Appendix Table A.2). The broad response pattern therefore shows that rival increases predict later Amazon increases.

Table 1. Raw Increase Probabilities After Rival Price Increases

Platform	P(Increase within 48h | No rival increase)	P(Increase within 48h | Rival increase)	Difference (p.p.)
Amazon	10.2	30.7	20.4
Walmart	8.5	13.5	4.9
Target	3.5	9.7	6.2
Best Buy	7.5	9.6	2.1
Home Depot	5.9	15.2	9.3

Notes: Entries report raw percentages in the main response-regression sample. The first column reports the probability that the focal platform increases its price within 48 hours when no rival increases price at time t. The second column reports the same probability after at least one rival selling the same product increases price at time t.

This broad pattern does not by itself show where the pricing sequence begins. A rival increase may already be a response to an earlier price increase by another platform. I therefore restrict attention to isolated rival increases, in which the ordering is unambiguous: no platform raised price for the product in the prior 48 hours, exactly one identified rival platform (for Amazon, any identified non-Amazon platform selling the product) then raises price, and the responding platform's own price is still unchanged at that moment.[6]

Table 2 establishes this response pattern in three steps. First, Amazon responds strongly to

[5] Among the response horizons I consider, the 48-hour window captures the largest share of later same-product adjustments, 33.9%.

[6] The analogous definition applies to isolated rival decreases.

Table 2. Responses to Isolated Rival Price Increases

	Price increase
Response to isolated rival price increase	
Amazon	12.94
	(1.79)
Non-Amazon major platforms	3.65
	(0.95)
Difference: Amazon − non-Amazon	9.29
	(1.73), $p < .001$
Observations	597,334
All-platform robustness	
All non-Amazon platforms	4.24
	(0.95)
Difference vs. Amazon	8.54
	(1.77), $p < .001$
Total observations	856,436
Response to isolated Amazon price increase	
Non-Amazon major platforms	0.15
	(0.73), $p = .837$
	$N = 406{,}877$
All non-Amazon platforms	−0.36
	(0.65), $p = .581$
	$N = 665{,}979$

Notes: Entries are percentage-point effects on increasing price within 48 hours after an isolated price increase, relative to otherwise comparable periods without one. The Amazon row uses isolated non-Amazon increases facing Amazon; the non-Amazon row pools Walmart, Target, Best Buy, and Home Depot when they face isolated increases by their own rivals. The difference row tests equality between these two coefficients, with its standard error, clustered by product, in parentheses. The all-platform robustness block repeats this comparison using every non-Amazon platform in the pricing panel, rather than only the four majors; the corresponding Amazon coefficient in that expanded sample is 12.78 percentage points with standard error 1.82. The bottom panel reverses direction, reporting non-Amazon platforms' response to an isolated Amazon price increase, first pooling only the four majors and then every non-Amazon platform in the panel; both coefficients are small and statistically indistinguishable from zero. All specifications include product, platform, week, and day-of-week fixed effects; standard errors are clustered by product.

rivals. After an isolated rival increase, Amazon is 12.9 percentage points more likely to raise its own price. Second, Amazon responds more strongly than other platforms generally do. The corresponding response among the four major non-Amazon platforms is only 3.7 percentage points, yielding a gap of 9.3 percentage points ($p < .001$). Expanding the comparison to all non-Amazon platforms leaves the conclusion unchanged: their pooled response is 4.2 percentage points, and the gap relative to Amazon remains 8.5 points.

Third, rivals do not similarly respond to Amazon. When Amazon is the isolated price increaser, the four major non-Amazon platforms show essentially no response: their response estimate is 0.15 percentage points ($p = .837$). Pooling all non-Amazon platforms gives a similarly small and statistically indistinguishable estimate of -0.36 percentage points ($p = .581$). Thus rival increases predict Amazon increases, but Amazon increases do not predict rival increases.

The same response-strength ranking appears for price decreases. Appendix Table A.1 shows that Amazon's response to isolated rival decreases is also much larger than other platforms' response: 12.8 percentage points versus 3.2 points. Unlike the increase case, however, isolated Amazon decreases also elicit detectable rival responses. The response asymmetry is therefore sharpest for upward price movements, while Amazon's unusually strong responsiveness appears in both directions.

These response patterns are difficult to reconcile with a pure delayed-common-shock explanation. First, if isolated rival increases mainly reflected product-level shocks that all platforms adjust to with a lag, the response should not be concentrated at Amazon. Panel B of Appendix Table A.2 shows that none of Walmart, Target, Best Buy, or Home Depot approaches Amazon's response individually. Second, if the initiating rival were merely the first platform to adjust to the same shock, Amazon should respond similarly across initiating rivals. Appendix Table A.3 shows otherwise: Amazon's upward response lift is 17.5 percentage points after isolated increases by major non-Amazon platforms, compared with 9.9 percentage points after increases by other platforms. This difference is not an artifact of the raw comparison: controlling for product, week, day of week, and the initiator's price-change size, Amazon's response is 15.7 percentage points higher after major-platform initiations (clustered standard error 5.8). The response margin is therefore tied not only to the existence of a rival price increase, but also to which rival moves first.

The isolated-event response is also stronger where the demand-side stakes are larger. The response estimate rises from 8.3 percentage points among low-demand products to 10.9 among middle-demand products to 15.5 among high-demand products (Appendix Table A.4). A continuous version of this test, interacting the treatment with log product demand, points in the same direction (1.9 percentage points per log-point of demand, $p = .057$). This pattern motivates the model below: when more demand is at stake, responding to a rival's price increase can be more valuable for a demand-advantaged platform.

The response pattern is not limited to products with few observed competitors. Appendix Table A.5 shows that Amazon's response lift is positive in every competitor-count bin and remains large when five or more rival platforms are observed for the same product.

Amazon's outsized response to rivals is also robust to the choice of response window. Restricting to isolated rival price increases, Amazon's response remains significantly larger than the pooled response of the other major platforms across alternative response windows: the gap is 4.4 percentage points at 12 hours ($p = .001$), 5.3 at 24 hours ($p = .001$), 9.3 at 48 hours ($p < .001$), and 12.0 at 72 hours ($p < .001$). The pattern also holds across product categories, as shown in Appendix Figure A.1.

2.4 Amazon Raises Price Soon After Rival Increases

Amazon is unusually likely to raise price after isolated rival increases. The next question is whether this pattern is tied to the rival's move. If Amazon were already on an upward path, the same events should predict Amazon increases before the rival move. If the estimate only reflected Amazon's frequent repricing, Amazon's increases should be spread throughout the response window rather than concentrated soon after the rival move.

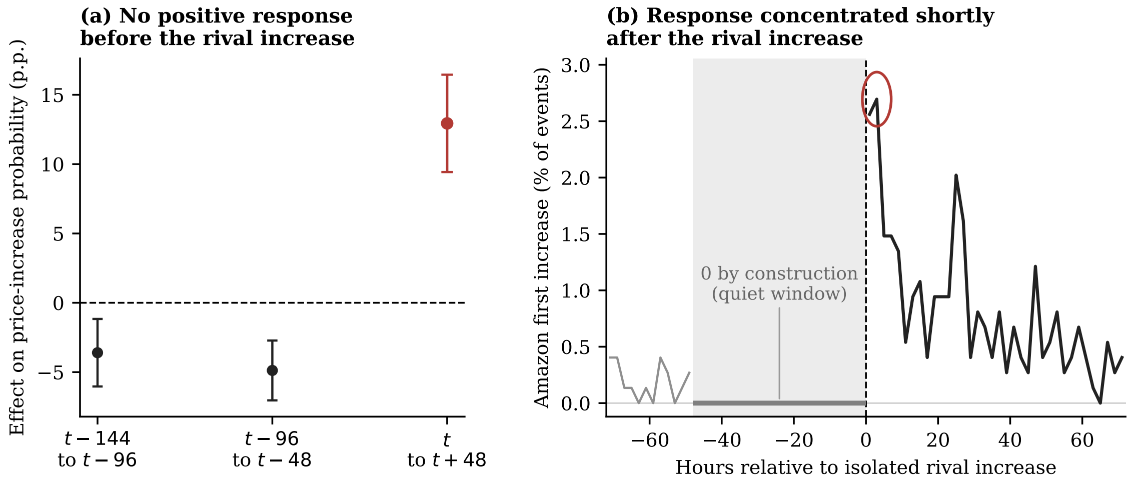

Figure 2. Timing Evidence Around Isolated Rival Price Increases.

Notes: Panel (a) reports Amazon's percentage-point effect on increasing price within 48 hours of an isolated rival increase. Placebo windows assign the same rival-increase events to fake event times 96 or 144 hours earlier. The placebo specifications include product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. Panel (b) conditions on isolated rival increases facing Amazon and plots the share of events for which Amazon's first subsequent price increase occurs in each two-hour bin; the vertical dashed line marks the isolated rival increase. The quiet window is zero by construction because the isolated event requires no same-direction price increase for the product during the prior 48 hours.

Figure 2 addresses both alternatives. Panel (a) shows no comparable positive response in the fake earlier windows. Panel (b) shows that Amazon's first increase is concentrated shortly after the rival move. Of Amazon's responses within the window, 28 percent occur within the first 8 hours, with no anticipatory movement beforehand. The timing evidence therefore supports the response interpretation: Amazon is not simply continuing a prior price increase or repricing often in general; it is raising price after the rival's observed increase.

2.5 Amazon Matches When Responding to Rival Price Increases

Amazon raises price soon after isolated rival increases. The next question is where Amazon's response lands. If Amazon matches the rival's new price, the higher price stands. If Amazon remains below the rival, Amazon's lower price limits the increase. I therefore measure Amazon's response price relative to the rival's new price.

I classify Amazon's first price increase within 48 hours as below, at parity with, or above the rival's new price. For example, if Walmart raises the price of a headphone from \$98 to \$109 and Amazon later moves from \$98 to \$109, I classify Amazon's adjustment as a parity response. If

Amazon moves to $108, I classify it as below the rival's new price; if it moves to $110, I classify it as above.

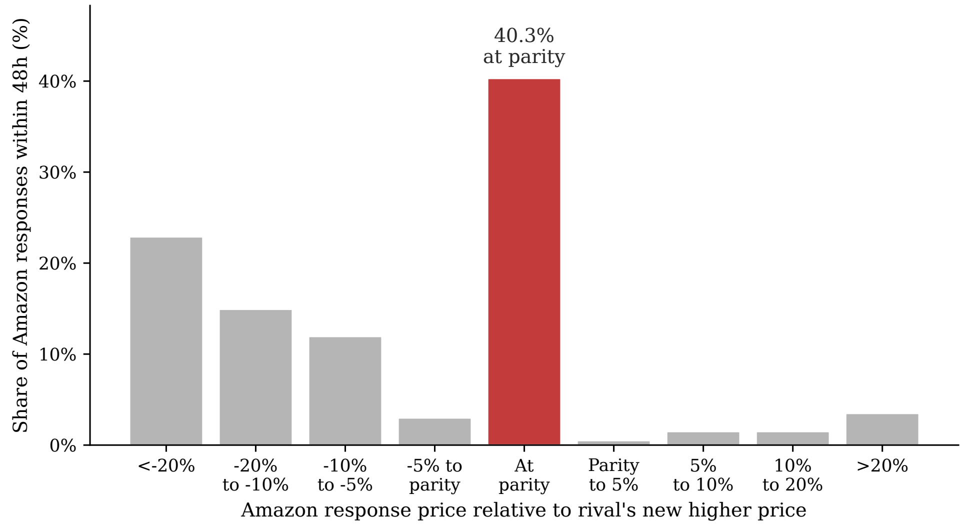

Figure 3. Amazon's Response Price Relative to the Rival's New Price.

Notes: The figure conditions on isolated rival price increases facing Amazon for which Amazon raises price within 48 hours. It plots Amazon's response price relative to the rival's new price. Gaps are measured in percentage points relative to the rival's new price; the parity bar includes responses classified as matches under the tolerance described in the text.

The largest mass in Figure 3 is at parity: 40.3% of Amazon's responses match the rival's new price. Including responses below the rival's new price, 93.0% are at or below the rival's new price. The important point is not price matching by itself, but the matching response of Amazon, the demand-advantaged platform, after a rival moves first. A substantial share of these responses preserve the rival's higher price rather than undoing it through undercutting.

The same tendency to land at parity also appears after rival price decreases: conditional on Amazon lowering price after an isolated rival decrease, 79.9% of responses match the rival's new price. The persistence analysis below asks whether this matching behavior has different consequences for increases and decreases.

2.6 Rival Increases Last Longer When Amazon Matches

I next ask whether rival increases last longer when Amazon matches them, and whether the same pattern holds for rival decreases. For each isolated rival price change facing Amazon, I compare two cases: Amazon matches the rival's new price within 48 hours, or Amazon does not match it within

48 hours. I then track whether the rival's price remains on the new side of the price change over the following 60 days. If Amazon's matching response helps preserve the rival's higher price, then rival increases followed by an Amazon match should last longer than rival increases not followed by an Amazon match.

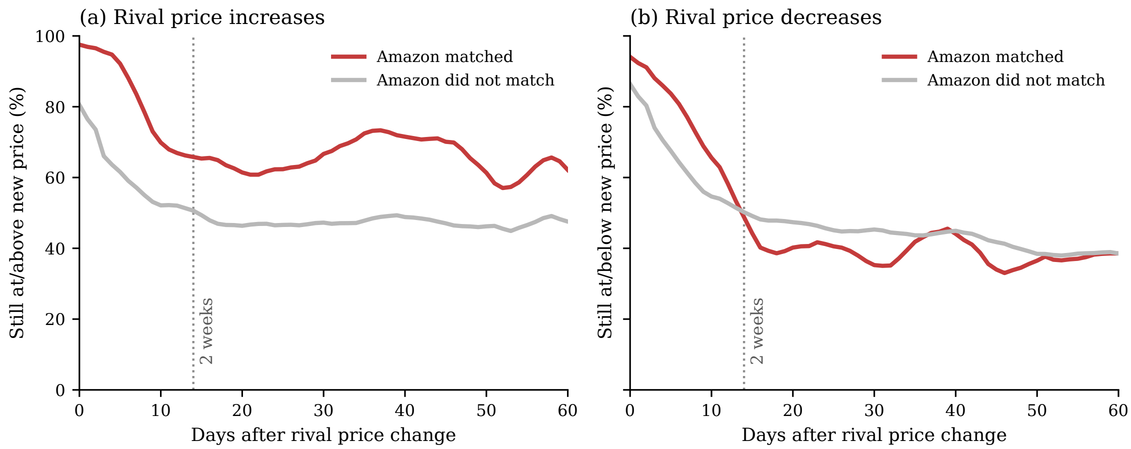

Figure 4. Persistence of Rival Price Changes by Amazon's Response.

Notes: The figure conditions on isolated rival price changes facing Amazon. Panel (a) compares rival increases that Amazon matches within 48 hours with rival increases that Amazon does not match within 48 hours. Panel (b) makes the analogous comparison for rival decreases. Lines report 5-day centered moving averages of daily persistence shares. For rival increases, persistence means that the rival's later price remains at least as high as the new price set immediately after the increase. For rival decreases, persistence means that the rival's later price remains no higher than the new price set immediately after the decrease. For example, a \$100-to-\$110 increase is counted as persistent only if the later rival price is at least \$110; a \$100-to-\$90 decrease is counted as persistent only if the later rival price is no higher than \$90.

The persistence gap is economically large and statistically significant. At two weeks, 67.1% of matched rival increases remain at or above the new price, compared with 51.3% of unmatched increases; the gap remains at 60 days (63.0% versus 47.1%). Product-clustered checkpoint tests show that the increase gap is statistically significant at two weeks (15.8 percentage points, $SE = 5.8$, $p = .007$) and remains positive at 60 days (15.9 percentage points, $SE = 8.3$, $p = .058$). In duration terms, matched rival increases also reverse later: the first reversal occurs after 25.5 days on average, compared with 17.3 days for unmatched increases, with duration capped at 60 days. Rival decreases show the opposite pattern: matched decreases persist less often than unmatched decreases at two weeks (44.6% versus 49.2%), though this gap narrows by 60 days (37.7% versus 38.6%). These decrease gaps are not statistically significant (-4.5 percentage points, $SE = 5.0$, $p = .361$ at two weeks; -0.9 percentage points, $SE = 7.2$, $p = .895$ at 60 days). Amazon's matching response is therefore associated with more durable higher rival prices specifically, not with durable rival price changes in general.

The content of Amazon's response also matters. Among isolated rival increases followed by an active Amazon response, increases followed by a parity response are 19.5 percentage points more likely to remain at or above the new price after two weeks than increases followed by an Amazon response that remains more than 5 percent below the rival's new price ($SE = 7.7$, $p = .013$). Thus, the persistence pattern is not driven by any Amazon price change; it is concentrated in responses that move Amazon to price parity.

The next section builds a model to explain why following can be incentive-compatible for a demand-advantaged firm.

3 Dominant Followership

When two undifferentiated firms compete only on price, the standard benchmark is the simultaneous-move Bertrand model, which predicts prices equal to marginal cost. Sequential timing alone does not overturn this intuition: if firms are otherwise symmetric and can undercut each other, the force pushing prices toward marginal cost remains. The standard intuition changes once the model is modified in a seemingly minor way: one firm has a demand advantage at price parity. Instead of assuming that firms are identical in every respect, I allow one firm, hereafter the dominant firm, to capture more than half of market demand when prices are equal. This assumption can be microfounded by a simple lexicographic decision rule. Consumers first compare prices, but when prices are equal or perceived as equivalent, they choose based on firm identity, brand, convenience, or other non-price advantages that favor the dominant firm. This advantage need not operate through the dominant firm charging high prices itself or moving first to claim the usual Stackelberg advantage. The question is instead how the dominant firm's response to a rival's price change shapes the resulting equilibrium.

The central result is that when the rival initiates and the dominant firm follows, a higher price can persist indefinitely as a Markov-perfect outcome. The analysis builds this result in three steps. First, in a one-shot pricing game, prices rise above the simultaneous-move level only when the demand-advantaged firm is the one who follows, since only then does the advantage turn the follower's incentive from undercutting to matching. Second, I add a period in which raising price is costly: the rival initiates only if the dominant firm's eventual match is valuable and likely, meaning the dominant firm faces low response friction, and only if the dominant firm goes last, since a rival that goes last instead would rather undercut the match than sustain it. Third, extending the game indefinitely, the pattern survives as a constructed Markov-perfect equilibrium when the dominant firm prefers waiting to raising the price on its own, and the rival prefers sustaining the match to undercutting it, both of which become easier when the rival faces stronger response friction.

3.1 Environment

Suppose there are two firms, indexed by $i \in \{D, S\}$. Firm D is the dominant, demand-advantaged firm, and firm S is the less demand-advantaged rival. Firms choose prices from a finite evenly spaced grid

$$\mathcal{P} = \{\delta, 2\delta, \ldots, M\delta\},$$

where $\delta > 0$ is the grid step and $M\delta$ is the maximum price. Let $p_i \in \mathcal{P}$ denote firm i's price. For any x, define the grid floor

$$\lfloor x \rfloor_{\mathcal{P}} = \max\{p \in \mathcal{P} : p \leq x\}.$$

For numerical illustrations, I normalize the maximum price to one and use a five-point grid, so $\delta = 0.2$ and $\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}$.

The grid reflects the discrete price changes observed in the data.[7]

Total demand is normalized to one. Demand is allocated according to

$$q_D(p_D, p_S) = \begin{cases} 1, & p_D < p_S, \\ \lambda, & p_D = p_S, \\ 0, & p_D > p_S, \end{cases} \qquad q_S(p_D, p_S) = 1 - q_D(p_D, p_S).$$

[7] It can also be interpreted as a reduced-form representation of limited consumer price perception: under Weber's law, the smallest noticeable change in a stimulus is proportional to its initial level, so sufficiently small relative price differences may generate the same perceived price and demand response (Fechner 1860; Monroe 1971, 1973). Observed pricing patterns are consistent with discrete, economically meaningful price steps: average dollar adjustment sizes are \$5.17 for products under \$50, \$13.08 for products between \$50 and \$150, \$28.55 for products between \$150 and \$500, and \$81.64 for products above \$500.

The low-price firm receives the entire market. When prices are equal, the dominant firm receives share $\lambda \in [1/2, 1]$, with $\lambda = 1/2$ corresponding to the symmetric benchmark. I treat λ as capturing non-price platform advantages, such as brand trust, membership programs, or switching frictions, rather than an advantage created within the model by current-period pricing.

This demand specification is deliberately stark because it is designed to reflect a distinctive feature of the data: Amazon's responses cluster at exact parity. After isolated rival price increases, Amazon's modal response is to land exactly on the rival's new price, accounting for 40.3 percent of cases, more than any other outcome (Figure 3). The same exact-parity pattern is even stronger after rival price decreases, where 79.9 percent of Amazon's responses match the rival's new price. Because exact parity is the most common response, the demand split at a zero price gap is directly payoff-relevant. The parameter λ captures the dominant firm's relative attractiveness among consumers choosing between platforms at the same price.

Timing and market clearing. The baseline analysis compares two standard one-shot timing benchmarks: simultaneous-move Bertrand competition, in which firms set prices without observing each other's current choice, and sequential-move Bertrand competition (Stackelberg timing), in which one firm, the leader, sets its price first and the other, the follower, observes it before responding. In both cases, the market clears only once, after both firms have set their prices, so there is no demand or payoff before the full price pair is determined. This one-shot structure implies the game ends with zero probability of either firm getting a further opportunity to respond. With marginal cost normalized to zero, realized profit is

$$\pi_i(p_D, p_S) = p_i q_i(p_D, p_S).$$

The two-clearing and infinite-horizon extensions below relax this zero-response-probability assumption, introducing timing friction as a force distinct from demand asymmetry.

3.2 Demand Advantage and Follower Incentives

Matching versus undercutting. The key mechanism can be seen from a firm's best response to a rival price $p \in \mathcal{P}$. On the price grid, the relevant local choice is whether to match p or undercut it by one step, $p-\delta$, since any lower price also receives the entire market but yields a lower margin. If the firm receives demand share s at price parity, matching yields sp, while undercutting yields $p-\delta$. The firm is therefore willing to match if and only if

$$sp \geq p - \delta,$$

or equivalently,

$$p \leq \frac{\delta}{1-s}. \tag{1}$$

The matching cutoff $\delta/(1-s)$ is increasing in the firm's demand share at parity.

Simultaneous pricing. Consider a candidate common-price equilibrium $p_D = p_S = p$ under simultaneous pricing. Applying condition (1), the dominant firm's no-undercutting condition is

$$p \leq \frac{\delta}{1-\lambda},$$

while the rival's condition is

$$p \leq \frac{\delta}{\lambda}.$$

Under demand asymmetry, $\lambda > 1/2$, the rival's condition is more restrictive. Hence the highest common-price equilibrium is

$$p^{Sim}(\lambda) = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

Thus, demand asymmetry by itself is not enough to raise prices in the simultaneous benchmark. The common price is disciplined by the rival, which has the stronger incentive to undercut.

Sequential pricing. Sequential pricing changes the constraint because the follower chooses its price after observing the first mover's price. In the one-clearing game, demand is realized immediately after the follower's response, so the first mover cannot respond again. The highest sustainable matched price is therefore pinned down by the follower's incentive to match rather than undercut, which depends on the asymmetric tie-breaking rule at price parity.

Let p_D^{follow} denote the highest matched price sustainable when the dominant firm follows, and let p_S^{follow} denote the corresponding price when the rival follows.

Proposition 1 (Dominant followership raises equilibrium prices)**.** *Suppose $\lambda > 1/2$. The highest matched price sustainable under sequential pricing satisfies*

$$p_D^{follow} \geq p_S^{follow}.$$

[8] *Specifically,*

$$p_D^{follow} = \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}},$$

while

$$p_S^{follow} = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

Role assignment and matched prices. The result follows from applying the no-undercutting condition (1) to the follower: $\lambda > 1/2$ implies $\delta/(1-\lambda) > \delta/\lambda$. Figure A.2 plots the comparative statics.

Table 3 illustrates the result in a five-point price grid.

Table 3. Highest Matched Equilibrium Prices on the Price Grid

Demand environment	Simultaneous pricing	Sequential $D \to S$	Sequential $S \to D$
Symmetric demand, $\lambda = 1/2$	2δ	2δ	2δ
Asymmetric demand, $\lambda = 0.7$	δ	δ	$\boxed{3\delta}$

Notes: Entries report the highest matched equilibrium price on the normalized five-point grid. Lower matched equilibria may also exist. Because both firms' payoffs are increasing in the matched price, the reported equilibrium is the payoff-dominant matched equilibrium within each timing environment. In sequential pricing, $D \to S$ denotes the dominant firm moving first and the rival following; $S \to D$ denotes the reverse.

When demand asymmetry raises prices. The table shows that the highest matched price rises above the simultaneous-move Nash equilibrium level of 2δ only when three things hold together: sequential timing, demand asymmetry, and the dominant firm playing follower rather than leader. Demand asymmetry alone is not enough: under simultaneous pricing or when the dominant firm leads ($D \to S$), the rival's no-undercutting condition binds either way, so the price is

[8] The inequality is weak because prices are chosen from a discrete grid; it is strict whenever $\mathcal{P}$ contains a price in $(\delta/\lambda, \delta/(1-\lambda)]$.

no higher than 2δ. Sequential timing alone is not enough either: without demand asymmetry, all three timing structures give a matched price of exactly 2δ. The price rises above 2δ only when asymmetric demand is paired with the dominant firm following ($S \to D$). In that case, the same price increase creates opposite response incentives: the dominant firm is willing to match because it keeps substantial demand at comparable prices, while the rival has a stronger incentive to undercut. Appendix B.2 formalizes this behavioral separation and shows that both firms can prefer the dominant-followership role assignment among matched sequential outcomes. Appendix B.6 shows that the same matching-versus-undercutting mechanism survives when firms retain loyal consumers at higher prices.

3.3 Costly Rival Initiation

The initiation problem. The one-clearing game abstracts from the short-run cost of raising price first: if the dominant firm matches, demand is realized only after both firms are at the higher price. I next ask whether rival initiation can still occur when the market clears once before the dominant firm responds. In that case, the rival may lose current demand before being matched. This setup lets me compare two orders that differ only in who holds the final response opportunity: the dominant firm or the rival.

Two-clearing timing. I write this timing as $i \to j \to M \to i \to M$, where M denotes a market clearing. Let a_1 denote firm i's initial price, a_2 firm j's response, and a_3 firm i's final price. The two market clearings occur at price pairs (a_1, a_2) and (a_3, a_2), so firm k's total payoff is

$$U_k(a_1, a_2, a_3) = \pi_k(a_1, a_2) + \beta\pi_k(a_3, a_2), \qquad k \in \{i, j\}, \tag{2}$$

where $\beta \in (0, 1]$ weights the second market. Firm i's final turn is a response window only if the rival moved away from the shared starting price, $a_2 \neq a_1$. Let $\rho_i \in [0, 1]$ denote the probability that this response window activates. If it activates, firm i chooses its response at a_3; otherwise, firm i's price remains at a_1. I treat ρ_i as an exogenous response-capacity parameter, not as a price-setting choice. The same response-window rule recurs, in its general repeated-game form, in the infinite-horizon and learning models below, illustrated in Figure 7.[9] I solve this finite game by backward induction.

Proposition 2 (Costly initiation depends on value, response probability, and order)**.** *Let* $p^M \equiv \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ *denote the highest price the dominant firm is willing to match.*

(i) Dominant firm responds last: $D \to S \to M \to D \to M$. *Fix the dominant firm's initial price at* δ.[10] *If* $\rho_D p^M > \delta$*, the rival raises price to* p^M *if and only if*

$$\beta > \bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D \, p^M - \delta}.$$

If $\rho_D p^M \leq \delta$*, no* $\beta \in (0, 1]$ *rationalizes initiation. The threshold is weakly decreasing in* λ *and in* ρ_D*; at* $\rho_D = 1$*, it reduces to* $\bar{\beta}(\lambda) = \delta/(p^M - \delta)$*.*

(ii) Rival responds last: $S \to D \to M \to S \to M$, with the rival's final response guaranteed ($\rho_S = 1$). *The higher price is never sustained. Because* $\lambda > 1/2$ *implies* $\lfloor \delta/\lambda \rfloor_{\mathcal{P}} = \delta$*, the rival is willing to match only at the grid floor. At any higher price, it prefers to undercut, so no discount factor* β *sustains a higher matched price under this order.*[11]

Corollary 3 in Appendix B.3 verifies that part (i)'s threshold is strong enough to support a complete equilibrium, not only a conditional response rule.

Two requirements for costly initiation. Proposition 2, part (i), gives two conditions for costly initiation. First, the future matched price must be high enough to compensate the rival for the demand it loses when it raises price first. Demand advantage helps by raising $p^M(\lambda)$, the highest price the dominant firm is willing to match. Second, the dominant firm's response must be likely enough: a match that occurs only rarely has little continuation value for the rival.

Figure 5 shows how these two conditions change the rival's initiation incentive. In the left panel, a larger demand advantage allows the dominant firm to profitably match a higher price, so the rival

[9] In this finite game, $a_2 \neq a_1$ is the analogue of a rival price change. The infinite-horizon game and the learning simulations use the same response-window rule: the window is triggered when the rival changes its posted price relative to its previous posted price.

[10] The grid's lowest price is a natural starting point for this illustration because it is the empirically relevant case: Amazon is typically at or near the lowest price among competitors (Figure 1: tied with the lowest rival in 54.2% of comparisons, priced above it in only 14.4%), though the grid floor here is a modeling device and not a claim that some absolute lowest price exists in practice.

[11] Appendix B.4 derives the equilibrium path when the rival responds last.

has more to gain from initiating. In the right panel, a higher dominant response probability makes that future match more likely. Both forces lower the patience required for the rival to raise price first: the rival is more willing to give up current demand when the future match is more valuable or more likely to occur.

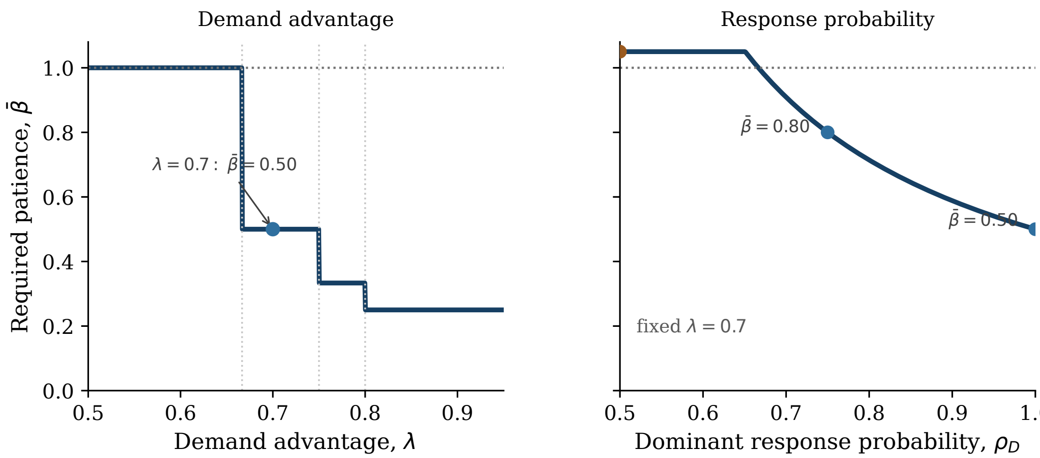

Figure 5. Costly Rival Initiation: Value, Response Probability, and Patience.

Notes: The figure uses the normalized five-point grid for the $D \to S \to M \to D \to M$ timing in Proposition 2 part (i), fixing the dominant firm's initial price at δ. Both panels plot $\bar{\beta}(\lambda, \rho_D) = \delta/(\rho_D\, p^M(\lambda) - \delta)$, the discount-factor threshold above which the rival raises price to $p^M(\lambda)$. The left panel fixes $\rho_D = 1$; dotted vertical lines mark the grid points at which $p^M(\lambda)$ steps up. The right panel fixes $\lambda = 0.7$, so $p^M(\lambda) = 3\delta$.

Who responds last. Who responds last matters, not only who moves first. When the dominant firm gets the final response, the rival can initiate from the low price and be matched. When the rival gets the final response, the same stable initiation path does not appear: the rival opens at the high price but later undercuts it. The reason is the response incentive from Proposition 1: with only minority demand at price parity, the rival does not want to preserve the high match when it can undercut instead. Table 4 illustrates this contrast.

Table 4. Equilibrium Price Path by Final Responder

Final responder	Move 1 (a_1)	Move 2 (a_2)	Move 3 (a_3)
Dominant firm, $D \to S \to M \to D \to M$	δ	3δ	3δ
Rival, $S \to D \to M \to S \to M$	3δ	3δ	2δ

Notes: Both rows use $\lambda = 0.7$ and $\beta = 0.7$. The first row corresponds to Corollary 3; the second corresponds to Appendix B.4.

3.4 Strategic Waiting and Persistence

Repeated-game temptations. The previous subsections establish the first two ingredients of dominant followership. First, demand advantage makes the dominant firm the natural matcher: because it keeps more demand at equal prices, matching a rival's higher price can be more profitable than undercutting it. Second, that response can make rival initiation worthwhile: the rival may be willing to raise price first if the future dominant match is valuable enough, which depends on demand advantage, and likely enough, which depends on the dominant firm's response probability. The infinite-horizon extension asks whether this response pattern remains sequentially rational once the game continues. Repetition creates two new temptations: before the rival initiates, the dominant firm could try to initiate the high price itself rather than wait; after the high price is matched, the rival could undo the match by undercutting.

Response-window state. Consider the two-price case $\mathcal{P} = \{L, H\}$, where $0 < L < H$. Firms alternate revision opportunities on a fixed clock, D then S then D then S. I use the same response-window rule as above. A firm moves normally unless its turn immediately follows a rival price change. In that case, it can revise only with probability ρ_i; otherwise prices remain unchanged for that period. The Markov state is (p_D, p_S, m, c): the current posted-price pair, the firm $m \in \{D, S\}$ whose turn is next, and an indicator c for whether the preceding period involved a price change.

Candidate policy. For the analytical result, I fix $\rho_D = 1$, so the dominant firm has no response friction, and vary ρ_S. The goal is not to classify all Markov-perfect equilibria, but to construct one stationary Markov equilibrium that implements the empirically documented pattern. I verify a candidate policy in which the rival initiates the high price from (L, L), the dominant firm matches, and both firms remain at (H, H):

price state (p_D, p_S)	$\sigma_D(p_D, p_S)$	$\sigma_S(p_D, p_S)$
(H, H)	H	H
(H, L)	L	H
(L, H)	H	H
(L, L)	L	H

where $\sigma_i(p_D, p_S)$ is firm i's price choice when it receives the revision opportunity.

Figure 6 shows the candidate path and the two repeated-game deviations that must be ruled out.

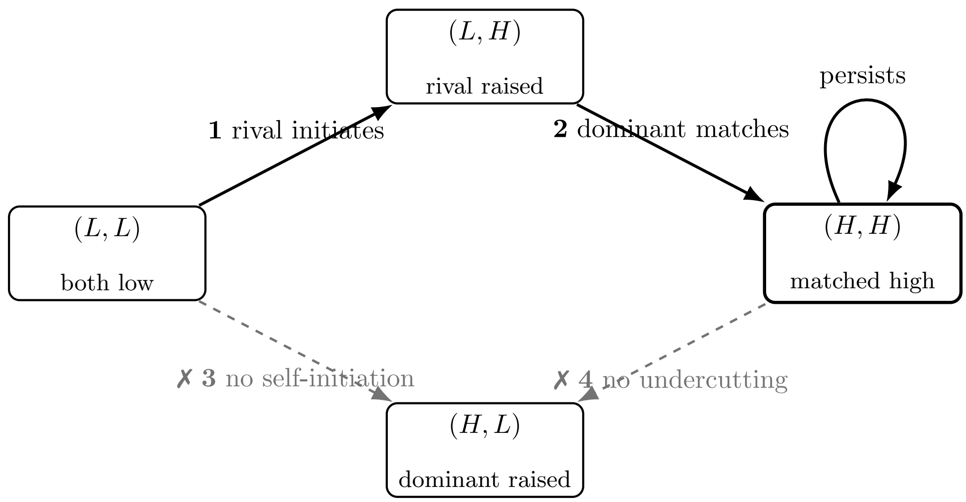

Figure 6. Candidate Dominant-Followership Policy

Notes: Solid arrows trace the candidate path: the rival initiates, the dominant firm matches, and play remains at (H,H). Dashed arrows mark the two repeated-game deviations that must be ruled out: dominant self-initiation from (L,L) and rival undercutting from (H,H).

What must be checked. Two checks carry over from the finite games. As in Proposition 2, the rival must be willing to initiate H from (L,L). As in Proposition 1, the dominant firm must be willing to match once the rival has raised price. The new repeated-game work is to rule out the two dashed deviations in Figure 6: (i) dominant self-initiation from (L,L), and (ii) rival undercutting from (H,H).

These two deviations share the same failure mode: both leave the game at (H,L), with the dominant firm high and the rival low, waiting on the rival's next response opportunity to resolve.

(i) *Dominant firm: self-initiation.* If the dominant firm raises from L to H on its own, it earns nothing at (H,L) until the rival's next response opportunity brings the rival's own price up to H as well. The dominant firm's gamble pays off only if the rival responds that way.

(ii) *Rival: undercutting.* If the rival instead undercuts from (H,H) down to L, it also lands at (H,L). But the candidate policy has the dominant firm revert to L there rather than remain at H, so only the rival's own next response opportunity can restore the high price it just gave up.

In both cases, the deviating firm's payoff hinges on how soon the rival can use its next response window, and that window activates less often when ρ_S is lower. Stronger rival response friction therefore weakens both temptations at once, sustaining the dominant-followership policy.

The proof solves the Bellman equations under this candidate policy and verifies these four deviations. Appendix B.5 gives the details.

Proposition 3 (Dominant followership can persist in repeated competition)**.** *Suppose* $\lambda > 1/2$, $\rho_D = 1$, *and*

$$(1-\lambda)H < L < \lambda H.$$

Let $x = L/H$, *and let* $\underline{x}(\lambda, \beta, \rho_S)$ *and* $\bar{x}(\lambda, \beta, \rho_S)$ *denote the no-self-initiation and no-undercutting thresholds derived in Appendix B.5. The candidate policy is a Markov-perfect equilibrium, and, starting from* (L, L)*, the market reaches* (H, H) *and remains there, whenever*

$$\underline{x}(\lambda, \beta, \rho_S) \leq x \leq \min\{\beta,\ \bar{x}(\lambda, \beta, \rho_S)\}.$$

This region is nonempty for a range of parameters satisfying the hypotheses above, and weakly expands as rival response friction rises, meaning ρ_S *falls, since* $\bar{x}$ *is weakly decreasing and* $\underline{x}$ *is weakly increasing in* ρ_S*.*

Together, Propositions 1–3 trace the full sequence of dominant followership: initiation, matching, and persistence. A high ρ_D governs initiation: raising price first means losing customers to the dominant firm until it responds, so the rival needs confidence that the dominant firm will respond with high probability (Proposition 2). Once the rival raises price, demand advantage, λ, is what makes that response a match rather than an undercut (Proposition 1). A low ρ_S then governs persistence: friction in the rival's subsequent response opportunities makes destabilizing deviations less attractive once the high price is reached (Proposition 3).

The model therefore uses response friction as a primitive difference across firms. In applications, ρ_i may reflect monitoring frequency, repricing-system reliability, engineering resources, or organizational constraints that make some firms more consistent responders than others. The equilibrium results do not require firms to choose these frictions inside the pricing game; they instead take pre-existing response capacities as given and show how those capacities shape pricing incentives.

The comparative static also shows that lower response capacity is not necessarily disadvantageous for the rival in this environment. When the rival can respond less frequently after the high price is matched, its temptation to undercut is weaker, making the high-price path easier to sustain. This is not a claim that firms choose ρ_i within the model; formalizing such a choice would require an earlier investment or commitment stage. Rather, lower response reliability need not be a friction the rival would always want to eliminate; in this environment, it can help the rival commit not to undercut the high matched price.

This interpretation motivates the learning treatment in Section 4, in which the demand-advantaged firm is also the low-friction responder: $\rho_D = 1$ and $\rho_S = 0$.

4 Algorithmic Learning with Dominant Followership

The analysis so far shows why dominant followership can be privately optimal when firms understand the incentives they face. In online retail, that is a strong assumption. Firms set prices across large product assortments, while demand conditions and rival response patterns vary across products and change in real time. A pricing algorithm may therefore have to learn the mechanism from market feedback rather than solve it from a known payoff model. I ask whether Q-learning algorithms can do so. Existing Q-learning models of algorithmic pricing typically study symmetric firms: either firms move simultaneously (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024) or they alternate sequentially with symmetric response opportunities (Klein 2021). The empirical evidence and the model point to a different learning environment. First, firms set prices sequentially: one platform's price change can become visible before a rival adjusts, and demand can be realized against the posted price in the interim. Second, demand advantage is itself asymmetric: Amazon holds a substantially larger share of demand at price parity than any single rival. Third, response friction is also asymmetric: Amazon appears to face lower friction in responding to rival price changes than other major platforms do. Section 3 shows how these asymmetries interact: demand advantage makes matching attractive for the dominant firm, a predictable dominant response makes rival initiation worthwhile, and limited rival response opportunities help the matched higher price persist. A learning environment that treated pricing as simultaneous, fixed demand shares symmetrically, or gave every firm the same response friction would erase exactly the mechanism these facts and this result identify. I therefore let demand advantage and response friction both vary and interact.

The results then proceed in three steps. I first ask when learned prices rise by isolating demand asymmetry, asymmetric response timing, and their combination. I then ask whose learning sustains the price increase by replacing one learner at a time with a myopic best response. Finally, I ask who gains from the learned price increase to see whether the learning that raises prices and the gains from those price increases accrue to the same firm. This distinction matters for interpreting the Project Nessie allegation, which turns on how learned price increases are generated and who benefits from them.

4.1 Environment and Learning Rule

Environment. I embed the demand environment from Section 3 in a repeated learning problem with two firms, D and S, both choosing prices from the same grid $\mathcal{P}$. Each firm's problem is a Markov decision process:

State. $s_{i,t} = (p_{i,t-1}, p_{j,t-1})$, $j \neq i$: the inherited posted-price pair firm i observes when it has a pricing opportunity. Appendix Section A.2 supports this two-price state empirically.

Action. $a_{i,t} \in \mathcal{P}$: the price firm i chooses from the common grid.

Reward. $\pi_i(p_{i,t}, p_{j,t})$: firm i's realized profit under the demand environment in Section 3.1, once both firms' posted prices are set for the period.

Timing. In the posted-price treatment, at most one firm adjusts its price in a period, and demand is realized against the resulting posted prices. The next turn then depends on whether the active firm changed its price. If firm i does not change price, firm j's scheduled turn proceeds normally. If firm i changes price, firm j's scheduled turn becomes a response window: with probability ρ_j, firm j can revise; with probability $1-\rho_j$, no firm moves and the market clears again at unchanged prices. Thus, ρ_j governs the response probability after rival price changes: lower ρ_j means stronger response friction. This differs from Klein (2021), where firms alternate deterministically with symmetric response opportunities. Here, ρ_D and ρ_S can differ, letting the simulations reflect the asymmetric response friction discussed above. Figure 7 summarizes the rule.

A separate simultaneous-updating benchmark shuts down this response channel entirely: both firms choose prices before demand is realized, so neither observes or responds to the rival's current move. I use it below to isolate the demand effect alone.

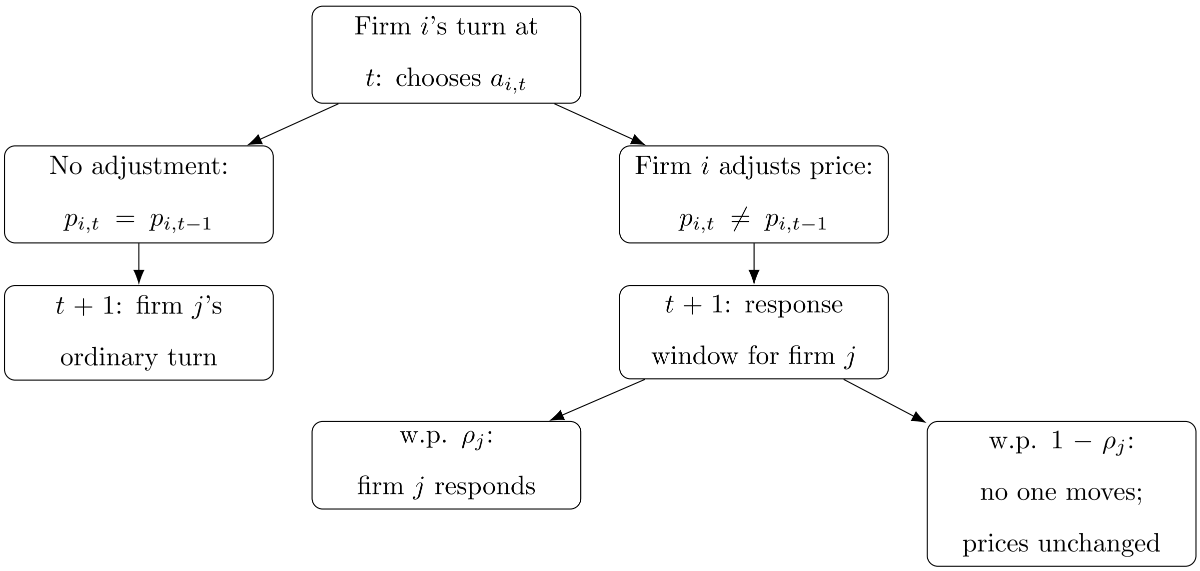

Figure 7. Response-Window Timing.

Notes: The figure illustrates the response-opportunity rule. After firm i's pricing opportunity at t, whether firm j's next scheduled turn is ordinary or gated depends on whether firm i changed its price. In a response window, firm j revises only with probability ρ_j.

Learning rule. Firms use standard Q-learning with ϵ-greedy exploration. Each firm maintains a table $Q_{i,t}(s, a)$, its estimate of the discounted value of price a in state s, and updates it each period as summarized in Algorithm 1.

Algorithm 1 Q-Learning Update for Firm i in Period t

Observe state $s_{i,t} = (p_{i,t-1}, p_{j,t-1})$

Draw $u_t \sim U[0, 1]$

if $u_t \leq 1 - \epsilon_t$ **then** ▷ exploit

  $a_{i,t} \leftarrow \arg\max_{a \in \mathcal{P}} Q_{i,t}(s_{i,t}, a)$

else ▷ explore

  $a_{i,t} \leftarrow$ price drawn uniformly at random from $\mathcal{P}$

end if

Post price $p_{i,t} \leftarrow a_{i,t}$

if simultaneous timing **then**

  $\widehat{Q}_{i,t} \leftarrow \pi_i(p_{i,t}, p_{j,t}) + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a')$

else ▷ response-window timing

  Accumulate discounted profit over periods $t, \ldots, t+m_i(t)-1$ while $p_{i,t}$ remains posted, as firm j does or does not respond (Figure 7)

  $\widehat{Q}_{i,t} \leftarrow$ accumulated profit + discounted continuation value ▷ two-turn case: Eq. (3)

end if

$Q_{i,t+1}(s_{i,t}, a_{i,t}) \leftarrow (1 - \alpha_t)\, Q_{i,t}(s_{i,t}, a_{i,t}) + \alpha_t\, \widehat{Q}_{i,t}$

The timing structure determines what the target $\widehat{Q}_{i,t}$ credits. In the simultaneous benchmark, the target is one period of profit plus continuation value:

$$\widehat{Q}^{sim}_{i,t} = \pi_i(p_{i,t}, p_{j,t}) + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a').$$

Under response-window timing, a chosen price remains active until the firm next receives a pricing opportunity. The target therefore credits the realized profit path while that price is posted, including profit earned after any rival response, plus continuation value from the resulting state. In the usual two-turn case,

$$\widehat{Q}^{rw}_{i,t} = \pi_i(p_{i,t}, p_{j,t}) + \beta\pi_i(p_{i,t+1}, p_{j,t+1}) + \beta^2 \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+2}, a'). \tag{3}$$

Appendix Section C.1 gives the exploration schedule, the exact response-opportunity rule, and the general target for cases in which a response window is not used.

4.2 Simulation Design

Main simulation treatments. The main simulations vary the two forces highlighted above: demand advantage and response friction. Firm D is fixed as the same firm across all four cells: the treatments turn its demand advantage and low-friction response advantage on or off, while firm S is its rival. Table 5 reports the resulting 2×2 design. In all four cells, both firms use Q-learning and prices follow the response-window timing described above. Only in the bottom-right cell does firm D hold both advantages; this is the dominant-followership treatment referenced throughout the results below. These treatments are the primary comparison for the price results in Table 7.

Table 5. Simulation Treatments

	Response timing	
Demand environment	Symmetric response	Asymmetric response
Symmetric demand	$\lambda = 0.5,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\lambda = 0.5,\ (\rho_D, \rho_S) = (1, 0)$
Asymmetric demand	$\lambda = 0.7,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\boxed{\lambda = 0.7,\ (\rho_D, \rho_S) = (1, 0)}$

Notes: In the symmetric-demand row, D and S split demand equally at price parity. In the asymmetric-demand row, D receives share $\lambda = 0.7$ at price parity. In the symmetric-response column, both firms have the same response probability. In the asymmetric-response column, D is the low-friction responder.

A separate simultaneous-updating benchmark shuts down the response channel entirely and compares $\lambda = 0.5$ with $\lambda = 0.7$, holding the price grid and learning rule fixed. This benchmark isolates the demand effect before introducing any response channel.

One-learner decomposition under dominant followership. The two-learner simulations show what happens when both firms adapt at the same time, where both demand asymmetry and response-friction asymmetry are present. Because both firms are learning simultaneously, however, this alone cannot reveal whose learning is necessary for the resulting price increase. To identify this, I hold the dominant-followership environment fixed, at $\lambda = 0.7$ and $(\rho_D, \rho_S) = (1, 0)$, and let one firm at a time follow a myopic best response instead of Q-learning. The myopic firm chooses the price that maximizes current-period profit against the rival's last posted price, without updating continuation values or internalizing how its choice affects future states.[12] Holding one side fixed also removes the moving-target problem created when two algorithms continually adapt to each other, giving the learning firm a stationary environment with a well-defined solution. Table 6 summarizes the three resulting cases; Table 10 below reports which one raises the market price.

Table 6. One-Learner Decomposition: Design (Dominant-Followership Environment)

	Firm S	
Firm D	Q-learning	Myopic best response
Q-learning	Both firms learn	Dominant firm only learns
Myopic best response	Rival only learns	*(excluded: no learning to identify)*

Notes: Each cell fixes one pricing rule per firm. The excluded cell is omitted because, with neither firm learning, there is no learning process left to decompose. The three remaining cells isolate whose learning drives the market price under dominant followership; the boxed cell is the two-learner baseline against which the other two cases are compared.

Outcome measurement. The main outcome is the transaction price, $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$, the price paid by consumers under the baseline demand allocation. I also report firm-level prices and profits to show how learning changes the distribution of gains. Unless otherwise stated, outcomes are averaged over the final 5 percent of periods across simulation runs.

4.3 When Do Learned Prices Rise?

Demand advantage alone does not raise learned prices. Only under dominant followership, when the demand-advantaged firm is also the one that responds reliably, does the price climb substantially above what firms learn without a response channel.

Simultaneous no-response benchmark. The simultaneous-updating benchmark removes the response channel and isolates the demand effect. The learned transaction price is 0.339 under symmetric demand and 0.274 under asymmetric demand: without a response channel, demand asymmetry lowers rather than raises learned prices, even in repeated learning. Appendix Figure A.4 shows the corresponding price trajectories.

[12] Ties between matching and undercutting are broken toward matching, consistent with the weak matching condition in equation (1).

Table 7. LEARNED TRANSACTION PRICES: DEMAND ADVANTAGE AND RESPONSE TIMING INTERACT

	Response timing	
	Symmetric response	Asymmetric response
Symmetric demand	0.388	0.533
Asymmetric demand	0.433	$\boxed{0.660}$

Notes: The table reports average transaction prices over the final 5 percent of periods in the two-learner Q-learning simulations. The transaction price is $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$.

Prices rise only when both asymmetries combine. Table 7 shows that the price increase requires the two asymmetries to coincide. Demand asymmetry alone raises the market price only modestly, from 0.388 to 0.433. Asymmetric response timing alone has a larger effect, raising the price to 0.533. The largest increase occurs under dominant followership, when the demand-advantaged firm is also the low-friction responder: the learned market price reaches 0.660, mirroring the mechanism in Section 3. Appendix C.3 shows that, under dominant followership, play concentrates increasingly in the high-price region over the course of training, even though the exact price pair continues to adjust.

Matching becomes far more likely under dominant followership. The key learning object is whether a higher-price attempt is rewarded or punished by the rival's response. Figure 8 measures this by asking whether firm D, which holds the demand and low-friction response advantages when those treatments are turned on, matches after firm S posts a price above D's inherited posted price.

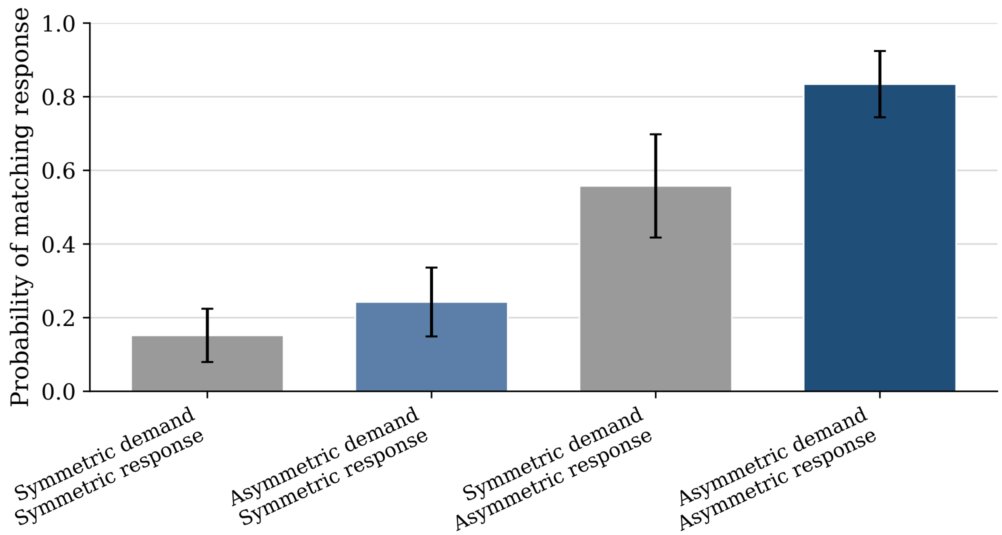

Figure 8. Firm D's Response to Rival Higher Prices.

Notes: Bars report the mean, across simulation runs, of the probability that firm D matches after firm S posts above D's inherited posted price, using events in the final 5 percent of periods. Whiskers report 95 percent confidence intervals across simulation runs.

The figure shows that matching becomes much more likely when the two asymmetries operate together. Demand asymmetry alone raises the matching probability modestly, from 0.15 to 0.24 under symmetric response. Response timing alone has a larger effect, raising it from 0.15 to 0.56 under symmetric demand. The largest effect occurs when the low-friction responder is also demand advantaged: the matching probability reaches 0.83. The dominant firm's response turns the rival's higher-price attempts from a risk into a reward. Appendix Figure A.6 decomposes the nonmatching responses.

Play concentrates in high-price states under dominant followership. Figure 9 shows where the algorithms spend time after learning, pooling posted-price pairs across simulation runs in the final part of training.

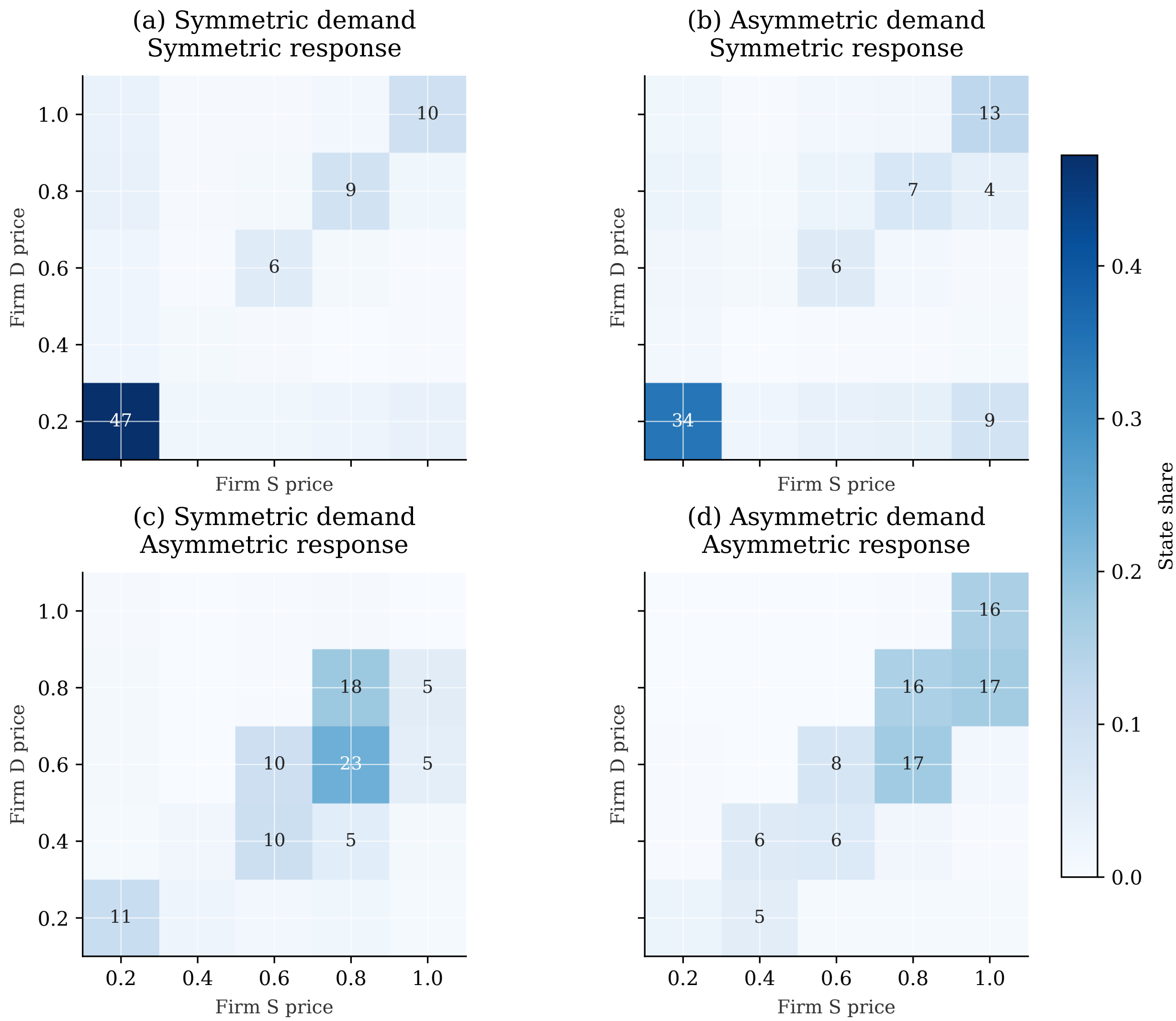

Figure 9. Visited Price States after Learning.

Notes: The figure pools posted-price pairs across all simulation runs in the final 5 percent of training. The vertical axis is firm D's posted price and the horizontal axis is firm S's posted price. Cell labels report percentage shares and are shown for cells with shares of at least 4 percent.

The heatmaps show that the high average prices in Table 7 are not driven by isolated price spikes. In the symmetric-demand, symmetric-response benchmark, nearly half of late-period observations remain at the lowest price pair. The two intermediate treatments show the same ranking as Table 7: the isolated demand effect leaves most mass concentrated near the lowest price pair, while the isolated response-timing effect shifts mass into mid-to-high price cells. Under dominant followership, both firms post high prices in most late-period observations. This matches the response feedback documented above: when higher-price attempts are repeatedly matched, late-period play settles into high-price states.

4.4 Misaligned Roles Break the Mechanism

The previous results leave one question unanswered: does it matter which firm has the high response probability, or only that one of them does? To find out, I swap roles: the dominant firm keeps its demand advantage, but the rival now gets the high response probability instead. If reliable responding alone is driving the price increase, the swap should not matter: prices should still rise. If dominant followership is the true mechanism, the swap should break it: a reliable responder raises prices only when it is also the firm that wants to match, and after the swap, it no longer is.

Reversing the responder role. I implement this counterfactual by fixing $\lambda = 0.7$, so demand advantage remains with firm D, but reversing the response probabilities: $(\rho_D, \rho_S) = (0, 1)$. In this misaligned treatment, the rival S, not the dominant firm D, is the firm with the high response probability.

Table 8. Aligned versus Misaligned Response Roles

	Aligned $(\rho_D, \rho_S) = (1, 0)$	Misaligned $(\rho_D, \rho_S) = (0, 1)$
Market price	0.660	0.398
Share of periods with a matched high price	0.393	0.021

Notes: Both columns fix $\lambda = 0.7$. The aligned column reproduces the dominant-followership treatment in Table 7. The misaligned column reverses which firm has the high response probability, holding the demand advantage fixed with firm D. Outcomes are averaged over the final 5 percent of periods across 30 simulation runs. A matched high price is a period in which both firms are tied at a price of 0.6 or above.

Misalignment does not merely fall short of the aligned outcome: the learned market price, 0.398, is below either single asymmetry alone (0.433 and 0.533 in Table 7). A high response probability raises prices only when it belongs to the demand-advantaged responder; when it belongs to the rival instead, the learned price increase disappears.

How misalignment breaks the learned mechanism. The difference comes from what each firm learns to do. I distinguish three types of upward moves. A *higher-price initiation* is an upward move that ends above the rival's pre-move posted price, creating a new higher target. An *exact match* ends at the rival's pre-move posted price. A *raise-below-rival* move raises the firm's own price but remains below the rival's pre-move posted price. Table 9 shows that role assignment determines which firm performs each task.

Table 9. Role Assignment and the Learned Mechanism

	Aligned	Misaligned
Upward moves by the rival, S		
Share that initiate a higher price	$\boxed{0.984}$	$\boxed{0.150}$
Share that exactly match the rival's price	0.015	0.138
Share that raise but remain below the rival	0.001	$\boxed{0.712}$
Upward moves by the dominant firm, D		
Share that initiate a higher price	0.030	0.932
Share that exactly match the rival's price	$\boxed{0.888}$	0.029
Share that raise but remain below the rival	0.082	0.039

Notes: Both columns fix $\lambda = 0.7$. A higher-price initiation is an upward move that ends above the rival's pre-move posted price. An exact match ends at the rival's pre-move posted price. A raise-below-rival move raises the firm's own price but remains below the rival's pre-move posted price. Entries are shares of the indicated firm's own upward moves (price increases) in the final 5 percent of periods across 30 simulation runs.

In the aligned treatment, the learned roles are sharply separated. When the rival S raises price, it almost always creates a higher target: 98.4 percent of S's upward moves are higher-price initiations. When the dominant firm D raises price, it usually matches the rival's posted price exactly: 88.8 percent of D's upward moves are exact matches. The rival tests a higher price, and the demand-advantaged firm repeatedly validates it by matching.

Misalignment breaks this division of labor in two ways. First, because $\rho_D = 0$ removes D's response opportunities after S changes price, D no longer plays the frequent matching role. Its upward moves are instead mostly higher-price initiations. Second, and more importantly, S's learning problem changes. Facing a dominant firm that no longer responds predictably, S's algorithm learns not to test higher prices: the share of S's upward moves that initiate a higher price falls from 98.4 percent to 15.0 percent. Instead, most of S's upward moves raise price while remaining below D's posted price. As a result, the market rarely reaches a matched high price: only 2.1 percent of periods are matched high-price states, compared with 39.3 percent in the aligned treatment.

4.5 Whose Learning Sustains Higher Prices?

Section 3 shows that the higher price is sustained by the dominant firm's willingness to match rather than undercut. One might therefore expect the dominant firm's algorithm to be the source of the price increase in the learning simulations. The one-learner benchmarks show the opposite. Prices rise when only the rival learns, not when only the dominant firm learns.

The asymmetry comes from how much each firm must discover for dominant followership to emerge through learning. The dominant firm observes the rival's higher price before choosing whether to match. The rival must raise price first, absorb the immediate market response, and learn whether the dominant firm later makes that higher price profitable. This mirrors Proposition 2's threshold for costly initiation: the rival must weight future profit heavily enough to justify the demand it sacrifices today; Q-learning offers one way to discover this weighting from repeated market feedback rather than from assumed rationality. The one-learner benchmarks isolate each firm's learning problem to identify which one is responsible for the price increase.

Table 10. Learned Prices by Learner Identity (Dominant-Followership Environment)

	Firm S	
Firm D	Q-learning	Myopic best response
Q-learning	$(0.667, 0.776)$	$(0.248, 0.232)$
Myopic best response	$\boxed{(0.592, 0.598)}$	*(excluded)*

Notes: Each cell reports (p_D, p_S), average posted prices over the final 5 percent of periods across 30 simulation runs, under the dominant-followership treatment ($\lambda = 0.7$, $\rho_D = 1$, $\rho_S = 0$). The market price is the period average of $\min\{p_{D,t}, p_{S,t}\}$, which need not equal $\min\{p_D, p_S\}$ computed from the cell averages shown here, since the lower-priced firm can vary across periods. When only one firm uses Q-learning, the other follows a myopic current-period best response; the excluded cell is omitted since, with neither learning, there is no learning process to decompose.

Table 10 shows that learning matters mainly on the initiation side. When only the dominant firm learns, the market price remains low, at 0.225. When only the rival learns, the market price rises to 0.589, close to the two-learner outcome of 0.660. The algorithm that drives the higher price is therefore not the dominant firm's algorithm, but the rival's.

4.6 Who Captures the Gains?

The rival's algorithm drives the price increase, but it does not capture most of the gains. This is the final asymmetry in the simulation results. Starting from the benchmark in which the dominant firm uses Q-learning and the rival follows a myopic best response, I let the rival switch to Q-learning. Figure 10 shows that this raises the market price by 0.435, from 0.225 to 0.660. Both firms' profits rise, but the gains are highly asymmetric: the dominant firm's profit increases by 0.397, compared with 0.039 for the rival.

Thus, the algorithm that raises prices and the firm that benefits most are not the same. The rival's learning makes higher prices easier to sustain, while the dominant firm's demand advantage allows it to capture most of the resulting surplus. This is the asymmetry behind the Project Nessie allegation described in the introduction: a dominant firm like Amazon need not run the sophisticated algorithm itself to exploit higher prices, so long as its rivals' algorithms learn to raise them.

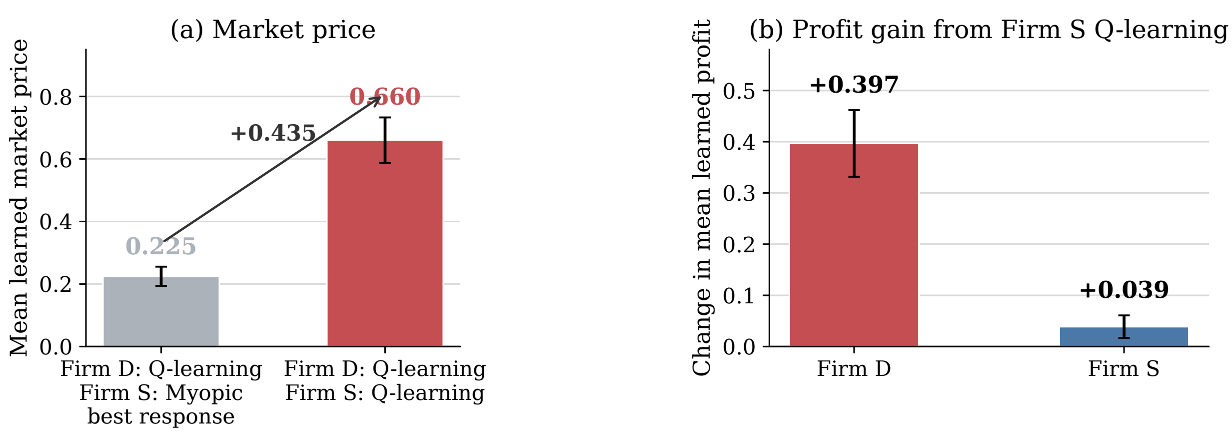

Figure 10. Market-Price and Profit Effects of the Rival's Q-Learning.

Notes: The figure uses the asymmetric-demand, asymmetric-response environment. It compares the benchmark in which the dominant firm uses Q-learning and the rival uses a myopic best response with the two-learner environment in which both firms use Q-learning. Panel (a) reports the change in the market price. Panel (b) reports changes in firm profits. Whiskers report 95 percent confidence intervals across 30 simulation runs. All outcomes are averaged over the final 5 percent of simulation periods.

Appendix C.5 shows that this asymmetric incidence holds at the run level as well: across simulation runs, those in which the rival posts a higher average price than the dominant firm also tend to be the runs in which the dominant firm earns the higher profit premium.

5 Conclusion

Can a dominant firm raise market prices through its responses to rivals' price changes? The central lesson of this paper is that it can: dominance can operate through response, not only through first moves. A demand-advantaged firm need not be the first mover, or even the firm whose algorithm learns to raise prices. If its response to rival increases is predictable, that response can change what rival algorithms learn to be profitable.

The evidence, model, and simulations identify different parts of the same mechanism. The data show the critical response pattern: Amazon often responds to and matches rival price increases, and matched increases are more durable. The model explains why this sequence can be privately optimal: the rival initiates, the dominant firm matches, and the higher price persists when the rival cannot easily reverse course. The simulations show how the same pattern can emerge from market feedback, with the rival's algorithm driving the price increase and the dominant firm capturing most of the gains.

This mechanism should not be evaluated only through price levels, or through whether firms explicitly coordinate: each response can be individually optimal given a firm's own demand position, without any agreement or common pricing rule. Yet once such responses become systematic and learnable, they can change what rival algorithms learn to be profitable, so a routine price response can become part of a feedback system that raises market-wide prices. Even without communication or agreement, such price increases can shift surplus from consumers toward firms through individually optimal responses that standard price-level screens may miss. The relevant object to measure is therefore the response system itself: whose price changes become visible first, who responds, how reliably, and whether the responder has a demand advantage. This is the same question raised by the Project Nessie allegation: counterfactual audits should ask whether market outcomes would change if any of these features of the response system were different.

This perspective changes how the absence of an Amazon price premium should be interpreted.

In a static price-level comparison, Amazon's position at or below rival prices looks like strong competition. In a sequential pricing environment, the same fact can reflect a different form of market power. When rivals move first, a dominant platform can match rather than undercut, remaining price-competitive in levels while shaping the prices that survive in the market. The competitive concern is therefore not only whether a dominant firm charges more than its rivals, but whether its responses make higher rival prices more likely to stand.

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A Empirical Appendix

A.1 Response Robustness and Heterogeneity

Table A.1. Responses to Isolated Rival Price Decreases

	Price decrease
Response to isolated rival price decrease	
Amazon	12.78
	(1.71)
Non-Amazon major platforms	3.17
	(0.92)
Difference: Amazon − non-Amazon	9.61
	(1.59), $p < .001$
Observations	597,334
All-platform robustness	
Amazon	12.69
	(1.72)
All non-Amazon platforms	3.88
	(0.77)
Difference vs. Amazon	8.81
	(1.65), $p < .001$
Total observations	856,436
Response to isolated Amazon price decrease	
Non-Amazon major platforms	2.84
	(0.95), $p = .003$
	$N = 406{,}877$
All non-Amazon platforms	3.45
	(0.80), $p < .001$
	$N = 665{,}979$

Notes: Entries are percentage-point effects on decreasing price within 48 hours after an isolated price decrease, relative to otherwise comparable periods without one. The table mirrors Table 2 for downward price movements. The Amazon row uses isolated non-Amazon decreases facing Amazon; the non-Amazon row pools Walmart, Target, Best Buy, and Home Depot when they face isolated decreases by their own rivals. The all-platform robustness block repeats this comparison using every non-Amazon platform in the pricing panel. The bottom panel reverses direction, reporting non-Amazon platforms' response to an isolated Amazon price decrease. Standard errors are reported in parentheses; for difference rows, the parenthesized value is the standard error of the difference. All specifications include product, platform, week, and day-of-week fixed effects; standard errors are clustered by product.

Table A.2. Price Adjustment Responses to Rival Price Changes

	Panel A: Any rival move		Panel B: Isolated rival move	
	Price increase	Price decrease	Price increase	Price decrease
Amazon	15.90	8.91	12.94	12.78
	(3.19), $p < .001$	(3.30), $p = .007$	(1.79), $p < .001$	(1.71), $p < .001$
Walmart	0.82	1.93	4.04	1.30
	(2.11), $p = .697$	(1.96), $p = .324$	(1.86), $p = .030$	(2.02), $p = .521$
Target	3.41	2.37	4.42	4.07
	(1.42), $p = .017$	(1.66), $p = .154$	(1.34), $p = .001$	(1.22), $p = .001$
Best Buy	-0.83	-0.94	1.93	1.47
	(1.29), $p = .521$	(1.53), $p = .539$	(1.34), $p = .149$	(1.21), $p = .224$
Home Depot	4.25	4.37	6.12	8.30
	(5.18), $p = .412$	(4.40), $p = .321$	(2.56), $p = .017$	(3.22), $p = .010$
Product FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Clustered SE	Product	Product	Product	Product
Observations	597,309		597,334	

Notes: The unit of observation is platform-product-time. The responding platform is required not to change price at t. Panel A uses any rival price move for the same product at t. Panel B restricts to isolated rival price moves, in which no platform raised price for the product in the prior 48 hours, exactly one identified rival platform moves at t, and the responding platform's own price is unchanged at that moment. Coefficients are percentage-point effects. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value.

Table A.3. Amazon Responses by Initiating Rival Type

Initiating rival	Events	Products	Response rate	Response lift
Major non-Amazon platforms	511	133	28.7%	17.5 pp
Other non-Amazon platforms	236	70	20.2%	9.9 pp
Major − other initiators, controlled			15.7 pp	(5.8)

Notes: The sample is Amazon observations after isolated non-Amazon upward first moves. Response rate is the probability that Amazon raises price within 48 hours. Response lift subtracts Amazon's baseline upward adjustment rate for products in the same initiating-rival group when that group did not initiate a clean upward first move. The controlled row estimates the major-versus-other difference within the treated sample, including product, week, day-of-week fixed effects and the initiator's percentage price increase; standard errors are clustered by product.

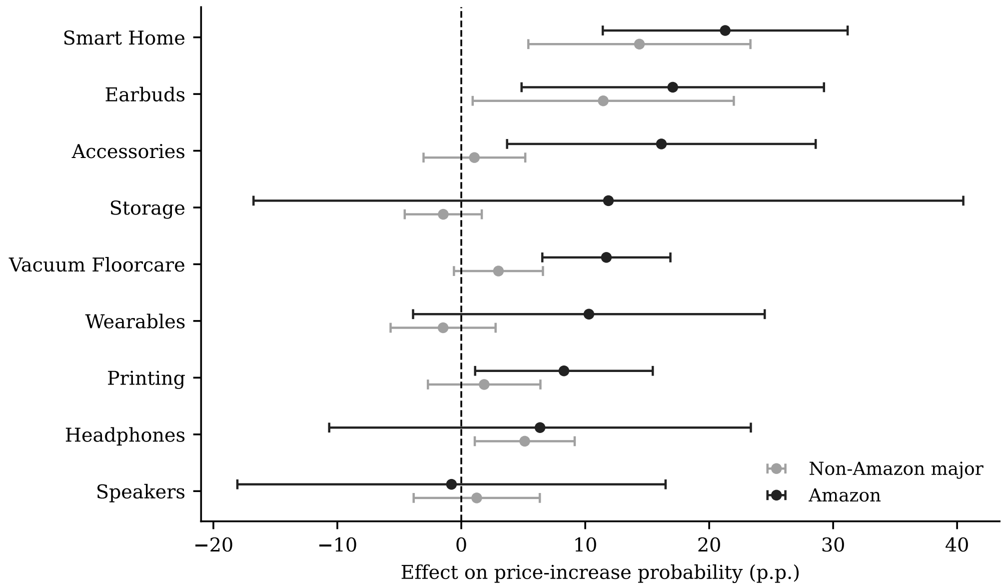

Figure A.1. Amazon Response to Isolated Rival Increases by Product Category.

Notes: The figure reports category-specific versions of the isolated-rival-increase response specification. Black markers report the Amazon coefficient; gray markers report the pooled coefficient for non-Amazon major platforms. Horizontal bars report 95 percent confidence intervals. Each specification includes product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. The figure includes categories with at least 20 Amazon-facing isolated rival increase events and at least five product clusters.

Table A.4. Amazon Response to Isolated Rival Increases by Product Demand Tier

	Low demand (50–2K)	Middle demand (2K–8K)	High demand (9K–100K)
Amazon	8.28	10.94	15.50
	(5.02), $p = .099$	(2.75), $p < .001$	(2.55), $p < .001$
Non-Amazon major platforms	2.35	2.71	4.36
	(3.00), $p = .435$	(1.28), $p = .034$	(1.27), $p < .001$
Products	59	62	67
Observations	144,795	190,916	255,024

Notes: Demand tiers are terciles of each product's median Amazon "Bought in Past Month" demand proxy, computed over the full matched-product panel; the ranges shown are the approximate floor-coded bands from Amazon's own display and are not exact boundaries. Entries are percentage-point effects on the probability of increasing price within 48 hours of an isolated rival price increase, estimated jointly for Amazon and the pooled non-Amazon major platforms (Walmart, Target, Best Buy, and Home Depot) within each tier. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value. All specifications include product, week, day-of-week, and platform fixed effects.

Table A.5. Amazon Response Lift by Number of Observed Rival Platforms

Competitor count	Events	Response rate	Baseline rate	Response lift
1–2 competitors	140	22.9%	10.3%	12.6 pp
3 competitors	108	30.6%	9.1%	21.4 pp
4 competitors	190	24.2%	10.1%	14.1 pp
5+ competitors	73	43.8%	12.9%	30.9 pp

Notes: Competitor count is the number of distinct non-Amazon platforms observed for the same product in the analysis panel. Events are isolated upward first moves by Walmart, Target, Best Buy, or Home Depot for which Amazon is observed at the initiating time. The response rate is the share of events followed by an Amazon price increase within 48 hours. The baseline rate is Amazon's probability of raising price within 48 hours in the same competitor-count bin when there is no clean upward first move by one of these major rivals. Response lift is the response rate minus the baseline rate.

A.2 Empirical State Dependence in Prices

The learning simulations use a parsimonious state: the platform's own lagged price and the lagged lowest rival price. Table A.6 supports this choice. Own prices are highly persistent, and the lagged lowest rival price helps explain the focal platform's current price even after conditioning on its own lagged price. The table also shows that a platform's position relative to the lowest rival price predicts the direction of its next adjustment: platforms priced above the rival benchmark tend to adjust downward, while platforms priced below it tend to adjust upward.

Table A.6. State-Space Motivation: Own and Rival Price States

	(1) $\log p_{ip,t}$	(2) $\log p_{ip,t}$	(3) $\Delta \log p_{ip,t}$
$\log p_{ip,t-1}$	0.9796***	0.9761***	
	(0.0046)	(0.0058)	
$\log p^{\min}_{-i,p,t-1}$		0.0080***	
		(0.0030)	
$\Delta \log p^{\min}_{-i,p,t-1}$			0.0105
			(0.0070)
$Gap_{ip,t-1}$			-0.0159***
			(0.0045)
Product FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Clustered SE	Product	Product	Product
Observations	878,992	878,992	849,328
R^2	0.999	0.999	0.008

Notes: The unit of observation is firm-product-time. The lag $t-1$ is the previous observed state for the same firm-product pair, restricted to observations between 1 and 6 hours apart. $p^{\min}_{-i,p,t}$ is the lowest rival price in the same product market, Δ denotes log price changes, and $Gap_{ip,t-1} = \log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}$. The sample restricts Amazon and Walmart to first-party listings where seller identity is observed. Standard errors are clustered by product. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Theoretical Appendix

B.1 Comparative Statics for Sequential Matched Prices

Figure A.2 plots the highest sustainable matched price as a function of demand asymmetry and the price-grid increment.

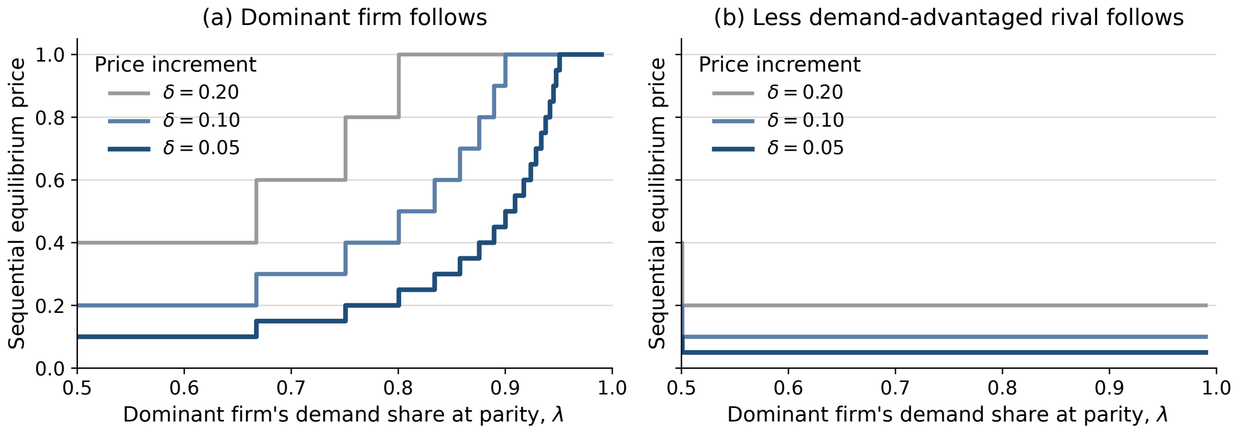

Figure A.2. Sustainable Matched Prices under Sequential Pricing.

Notes: Panel (a) shows the case in which the dominant firm follows, while Panel (b) shows the case in which the rival follows. Prices are the highest sustainable matched prices under the relevant follower constraint. Different lines correspond to different price-grid increments δ; smaller δ gives a finer price grid, so equilibrium prices adjust in smaller steps.

B.2 Behavioral Separation and Role Preference

Proposition 1 has two implications. First, there is a price region in which the dominant follower prefers matching while the rival follower prefers undercutting, so the same higher price can be sustained only when the dominant firm follows. Second, among matched sequential outcomes, both firms can prefer the role assignment in which the demand-advantaged firm follows.

Corollary 1 (Behavioral separation under demand asymmetry)**.** *Suppose $\lambda > 1/2$. For any first-mover price $p \in \mathcal{P}$ satisfying*

$$\frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda},$$

the dominant firm matches as the follower, while the rival undercuts.

Let $\Delta_i^U(p)$ denote firm i's net gain, as follower, from undercutting a first-mover price p, relative to matching. For the dominant firm,

$$\Delta_D^U(p) = (1-\lambda)(p-\delta) - \lambda\delta = (1-\lambda)p - \delta.$$

For the rival,

$$\Delta_S^U(p) = \lambda(p-\delta) - (1-\lambda)\delta = \lambda p - \delta.$$

The incentives have opposite signs exactly when

$$\Delta_D^U(p) < 0 < \Delta_S^U(p) \quad \Longleftrightarrow \quad \frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda}.$$

Corollary 2 (Role preference among matched outcomes)**.** *Among matched sequential outcomes, both firms weakly prefer $S \to D$ to $D \to S$.*

Let $S \to D$ denote the outcome in which the rival moves first and the dominant firm follows, and let $D \to S$ denote the reversed role assignment. Under $S \to D$, the matched price is p_D^{follow}; under $D \to S$, it is p_S^{follow}. At any matched price p, payoffs are

$$\Pi_D(p) = \lambda p, \qquad \Pi_S(p) = (1-\lambda)p.$$

Thus, $\Pi_i^{S\to D} = \Pi_i(p_D^{follow})$ and $\Pi_i^{D\to S} = \Pi_i(p_S^{follow})$. Since Proposition 1 gives $p_D^{follow} \geq p_S^{follow}$, both firms earn weakly higher payoffs under $S \to D$ among matched outcomes.

B.3 Proof of Proposition 2 and Corollary 3

Corollary 3 (A verified equilibrium)**.** *On the normalized five-point grid, for $\lambda \in (2/3, 3/4)$, $p^M = 3\delta$ and $\bar{\beta} = 1/2$. In this region, for $\beta > 1/2$, δ is also the dominant firm's own optimal initial price, so the unique equilibrium path is*

$$\delta \to 3\delta \to 3\delta.$$

The game is solved by backward induction.

The dominant firm's final response. The dominant firm's final response solves

$$a_3^*(a_2) \in \arg\max_{a_3 \in \mathcal{P}} \pi_D(a_3, a_2).$$

The discount factor does not affect this final choice, because only the second market payoff remains. Given a rival price p, matching yields λp, while undercutting by one grid step yields $p - \delta$. Thus the dominant firm weakly prefers to match p if and only if

$$\lambda p \geq p - \delta,$$

or $p \leq \delta/(1-\lambda)$. Let

$$p^M \equiv \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}}$$

denote the highest grid price the rival can induce the dominant firm to match.

The rival's response threshold (part (i)). Suppose the dominant firm initially posts δ, and suppose $\rho_D p^M > \delta$. The rival's response solves

$$a_2^*(a_1) \in \arg\max_{a_2 \in \mathcal{P}} \left\{ \pi_S(a_1, a_2) + \beta \pi_S\big(a_3^*(a_2), a_2\big) \right\}.$$

If the rival ties at δ, the lowest grid price, then $a_2 = a_1$, so the dominant firm's final turn is ordinary and ρ_D does not enter: the dominant firm has no reason to move away from the tie, so the rival's payoff from tying is $(1+\beta)(1-\lambda)\delta$. If the rival instead raises price to p^M, then $a_2 \neq a_1$, so the dominant firm's final turn is a response window: the rival is matched only when that window activates, giving expected payoff $\beta\rho_D(1-\lambda)p^M$. Lower price increases yield a smaller matched continuation price, while prices above p^M are not matched. The rival therefore prefers raising price to p^M whenever

$$\beta\rho_D(1-\lambda)p^M > (1+\beta)(1-\lambda)\delta \quad \Longleftrightarrow \quad \beta > \bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D\, p^M - \delta},$$

which reduces to $\bar{\beta}(\lambda) \equiv \delta/(p^M - \delta)$ when $\rho_D = 1$. If instead $\rho_D p^M \leq \delta$, the left side is non-positive for any $\beta > 0$ while the right side is strictly positive, so the inequality fails for every β; the division above is invalid in this case, and no discount factor rationalizes initiation. Since $p^M = \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ is a step function that weakly increases in λ, and the closed form is decreasing in ρ_D directly, the threshold $\bar{\beta}(\lambda, \rho_D)$ is weakly decreasing in both λ and ρ_D.

Reverse order (part (ii)). Part (ii) is proved in Appendix B.4, which derives the rival's final-turn incentive directly under the order $S \to D \to M \to S \to M$.

Verifying the equilibrium for a specific region. The threshold $\bar{\beta}(\lambda)$ is conditional on the dominant firm posting δ. It remains to verify that δ is the dominant firm's own best initial choice. Consider $\lambda \in (2/3, 3/4)$, so that $\delta/(1-\lambda)$ lies between 3δ and 4δ and $p^M = 3\delta$. Then $\bar{\beta} = 1/2$, so for $\beta > 1/2$ the rival's best response to an initial dominant price of δ is

$$a_2^*(\delta) = 3\delta.$$

The rival sacrifices current demand to create a higher matched continuation state.

Applying the same response problem to the other possible initial dominant prices, for $\beta > 1/2$ in this region, gives

$$a_2^*(2\delta) = \delta, \quad a_2^*(3\delta) = 2\delta, \quad a_2^*(4\delta) = 3\delta, \quad a_2^*(5\delta) = 4\delta.$$

Finally, the dominant firm's initial choice solves

$$a_1^* \in \arg\max_{a_1 \in \mathcal{P}} \left\{ \pi_D\big(a_1, a_2^*(a_1)\big) + \beta \pi_D\big(a_3^*(a_2^*(a_1)), a_2^*(a_1)\big) \right\}.$$

Given the response policy above, choosing δ gives the dominant firm current market demand and induces the rival to choose 3δ, which the dominant firm then matches. The resulting payoff, $\delta + 3\beta\lambda\delta$, strictly exceeds the payoff from every other initial price in this region. Hence, the unique equilibrium path is

$$\delta \to 3\delta \to 3\delta.$$

B.4 Reverse Timing in the Two-Clearing Game

This appendix proves part (ii) of Proposition 2. Consider instead the order $S \to D \to M \to S \to M$, so that the rival receives the final revision opportunity. The rival's final response is the same follower problem behind p_S^{follow} in Proposition 1: it matches p if and only if $p \leq \delta/\lambda$. Since $\lambda > 1/2$, $\delta/\lambda < 2\delta$, so the rival's matching set never extends above the lowest grid price:

$$a_3^*(p) = \begin{cases} \delta, & p = \delta, \\ p - \delta, & p > \delta. \end{cases}$$

Consequently, the price pair $(a_2, a_3^*(a_2))$ at the second clearing is never a tie above δ: it is undercut whenever $a_2 > \delta$, and tied only at δ itself. This holds regardless of what the dominant firm is induced to accept at the first clearing, so no discount factor β sustains a matched price above δ under this order.

For example, at $\lambda \in (2/3, 3/4)$ and $\beta < 3(1-\lambda)$, backward induction gives the equilibrium path

$$3\delta \to 3\delta \to 2\delta :$$

the rival's own optimal opening price is 3δ, the dominant firm matches it at the first clearing, and the rival then undercuts its own initiated price at the final revision. This is not the only pattern: for other values of λ, the rival's optimal opening price can instead exceed p^M, in which case the dominant firm undercuts rather than matches at the first clearing, and no tie above δ forms at all. Either way, the second clearing never sustains a match above δ.

B.5 Proof of Proposition 3

Fix the candidate policy in Section 3.4. The proof verifies this policy by solving its continuation values and checking one-period deviations. Once the policy is fixed, each state has a value: current profit from the prescribed price choice plus the discounted value of the next state generated by the revision process. Thus, for firm i, the value of state (p_D, p_S, m, c) satisfies

$$V_i(p_D, p_S, m, c) = \pi_i\big(p'_D(\sigma_m), p'_S(\sigma_m)\big) + \beta\mathbb{E}\left[V_i(p'_D(\sigma_m), p'_S(\sigma_m), m', c')\right],$$

where only the active firm's price can change before the market clears, m' is the next scheduled mover, and c' records whether the active firm changed price. The expectation is over whether a response window activates. These Bellman equations are linear because the policy is fixed. I then check one-period deviations. Since there are only two prices, each deviation replaces the active firm's prescribed price with the other price and compares the resulting payoff with the value of following the policy.

Let $x = L/H$. Solving the Bellman system and checking one-period deviations gives four sufficient conditions, each corresponding to one of the economic forces in the main text.

Rival initiation. At (L, L), the rival must prefer raising to H to staying at L. This gives

$$x \leq \beta.$$

The rival gives up current demand when it raises first, so initiation requires enough weight on the future matched price.

Dominant matching. At (L, H), the dominant firm must prefer matching H to undercutting at L. This gives

$$x \leq \lambda.$$

Matching yields λH, while undercutting yields L, so demand advantage makes matching optimal when $L/H \leq \lambda$.

No dominant self-initiation. At (L, L), the dominant firm must prefer staying at L to raising first to H. This is the formal version of the waiting logic: by waiting, the dominant firm earns low-price demand and lets the rival bear the cost of initiating. In the pure-alternation case, this comparison reduces to

$$\lambda L + \beta L \geq \beta \lambda H,$$

or $\beta\lambda/(\lambda + \beta) \leq x$.

No rival undercutting. At (H, H), the rival must prefer staying at H to undercutting to L. This is the persistence condition: undercutting gives the rival a current gain, but it must then work back to the high matched price. In the pure-alternation case, this comparison reduces to

$$\frac{(1-\lambda)H}{1-\beta} \geq L + \beta(1-\lambda)L + \beta^3 \frac{(1-\lambda)H}{1-\beta},$$

or $x \leq (1-\lambda)(1+\beta+\beta^2)/(1-\beta\lambda+\beta)$.

For general ρ_S, the last two conditions depend on the continuation value of the off-path state (H, L). I therefore solve these checks directly from the Bellman system. The pure-alternation case $\rho_S = 1$ gives the transparent benchmark

$$\frac{\beta\lambda}{\lambda+\beta} \leq x \qquad \text{and} \qquad x \leq \frac{(1-\lambda)(1+\beta+\beta^2)}{1-\beta\lambda+\beta}.$$

When $\rho_S < 1$, the same Bellman system accounts for the possibility that the rival cannot respond immediately after a price change. This lowers the continuation value of deviations that pass through (H, L). For the rival's undercutting deviation, the resulting threshold is

$$x \leq \bar{x}(\lambda, \beta, \rho_S) \equiv \frac{(1-\lambda)\left[1 - \beta^3 \rho_S - \beta^5 (1-\rho_S)\right]}{(1-\beta)\left[1 + (1-\lambda)\big(\beta + (1+\beta)\beta^2(1-\rho_S)\big)\right]},$$

which reduces to the $\rho_S = 1$ threshold above and is weakly decreasing in ρ_S whenever the rival-initiation condition $x \leq \beta$ holds.

The dominant firm's self-initiation deviation follows the same logic, with one added wrinkle. After the dominant firm's own deviation, the rival's first gated opportunity to complete the match may be missed; but here, the dominant firm's own policy then reverts the price back down, and that reversion is itself a price change that re-gates a second attempt for the rival before the match resolves automatically. With two sequential gated attempts instead of one, the deviation payoff is a probability-weighted average over three recovery paths, matched on the first attempt, matched on the second, or resolved automatically once both are missed. This makes the resulting threshold, denoted $\underline{x}(\lambda, \beta, \rho_S)$, quadratic rather than affine in ρ_S. It reduces to the $\rho_S = 1$ threshold above and is weakly increasing in ρ_S: stronger rival response friction makes it less likely that the rival completes the match quickly, so self-initiation pays off less.

For example, at $(\lambda, \beta, \rho_D, \rho_S, x) = (0.7, 0.9, 1, 0, 0.65)$, solving the Bellman system gives strictly positive gaps for all four deviations. By continuity, the candidate policy is therefore optimal on a nonempty neighborhood of this point. Starting from (L, L), once the rival receives a revision opportunity it chooses H, the dominant firm then matches, and the market converges to (H, H).

Both conditions therefore weakly expand the set of parameters that sustain dominant followership as ρ_S falls, which is why stronger rival response friction makes the candidate policy easier to sustain.

B.6 Loyal-Demand Extension

The baseline model isolates the matching-versus-undercutting mechanism by assuming that a firm charging above its rival receives no demand. This extension relaxes that assumption by allowing each firm to retain loyal consumers even when it charges more than its rival. With loyal demand, the dominant follower can monetize demand advantage in two ways: it can match the rival's price, or it can remain above the rival and serve only loyal consumers. Matching must therefore beat not only undercutting, but also staying expensive.

Proposition 4 (Dominant followership with loyal demand)**.** *Fix a leader price $p = k\delta$, with $k \in \{2, \ldots, M-1\}$. In the loyal-switcher environment, there exists a nonempty open set of primitives (α, β, λ), with*

$$0 < \beta < \alpha, \qquad \alpha + \beta < 1, \qquad \lambda \in [1/2, 1),$$

such that the dominant follower's unique best response is to match p, while the rival's unique best response is to undercut p by one grid step. Thus, the dominant-followership response pattern can arise even when both firms retain positive demand above the leader's price.

Figure A.3 shows that this can hold on a nonempty parameter region. Loyal demand narrows the mechanism, but does not eliminate it: when the dominant firm still gains enough from matching and the rival remains sufficiently dependent on switchers, the behavioral separation behind dominant followership survives.

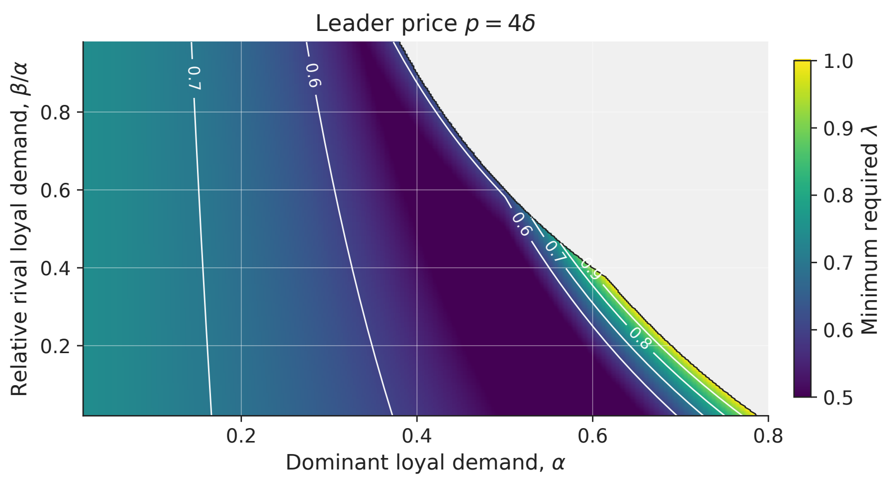

Figure A.3. Dominant Followership with Loyal Demand.

Notes: The figure fixes $M = 5$ and leader price $p = 4\delta$. It plots the minimum λ required for the dominant follower to match while the rival undercuts. Colored regions satisfy $\bar{\lambda}(\alpha, \beta, 4, 5) < 1$; gray regions admit no feasible $\lambda < 1$.

Proof of Proposition 4. Fix a leader price $p = k\delta$, with $k \in \{2, \ldots, M-1\}$. Total demand is normalized to one. A mass α is loyal to the dominant firm, a mass β is loyal to the rival, and the remaining mass

$$s = 1 - \alpha - \beta$$

consists of switchers, with $0 < \beta < \alpha$ and $\alpha + \beta < 1$. Loyal consumers buy from their preferred firm regardless of relative prices. Switchers buy from the lower-price firm, and at equal prices the dominant firm receives share $\lambda \in [1/2, 1]$ of switchers. Demand for the dominant firm is therefore

$$q_D(p_D, p_S) = \begin{cases} \alpha + s, & p_D < p_S, \\ \alpha + \lambda s, & p_D = p_S, \\ \alpha, & p_D > p_S, \end{cases}$$

while demand for the rival is

$$q_S(p_D, p_S) = \begin{cases} \beta, & p_D < p_S, \\ \beta + (1-\lambda)s, & p_D = p_S, \\ \beta + s, & p_D > p_S. \end{cases}$$

The proof first reduces the follower's response problem to three relevant actions, then derives the response inequalities, and finally shows that the feasible set is nonempty.

Step 1: Relevant follower actions. For either follower, matching gives demand at price parity. Among prices below the leader, $p - \delta$ is optimal because any lower price receives the same demand at a lower margin. Among prices above the leader, $M\delta$ is optimal because the firm receives only loyal demand and profit is increasing in price. Thus, it is enough to compare three actions: match, undercut by one grid step, and choose the loyal-only high-price option.

For the dominant follower, the three payoffs are

$$\pi_D^{match} = k\delta[\alpha + \lambda s], \qquad \pi_D^{under} = (k-1)\delta(1-\beta), \qquad \pi_D^{high} = M\delta\alpha.$$

For the rival, they are

$$\pi_S^{match} = k\delta[\beta + (1-\lambda)s], \qquad \pi_S^{under} = (k-1)\delta(1-\alpha), \qquad \pi_S^{high} = M\delta\beta.$$

Step 2: Response inequalities. The dominant follower matches if matching beats both deviations:

$$\pi_D^{match} > \max\{\pi_D^{under}, \pi_D^{high}\}.$$

Substituting the payoffs gives

$$\lambda > \max\left\{\frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}\right\}.$$

The rival undercuts if undercutting beats both alternatives:

$$\pi_S^{under} > \max\{\pi_S^{match}, \pi_S^{high}\},$$

or equivalently,

$$\lambda > \frac{1-\alpha}{ks}$$

and

$$(k-1)(1-\alpha) > M\beta.$$

The first inequality makes undercutting better than matching; the second makes undercutting better than serving only loyal consumers at the high price.

Step 3: Feasibility. Combining the λ-restrictions gives the cutoff

$$\bar{\lambda}(\alpha, \beta, k, M) \equiv \max\left\{\frac{1}{2}, \frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}, \frac{1-\alpha}{ks}\right\},$$

where $s = 1 - \alpha - \beta$. The dominant-followership response pattern holds whenever

$$\lambda > \bar{\lambda}(\alpha, \beta, k, M)$$

and

$$(k-1)(1-\alpha) > M\beta.$$

For fixed (α, β, k, M), an admissible $\lambda \in [1/2, 1)$ therefore exists if $\bar{\lambda}(\alpha, \beta, k, M) < 1$ and the additional condition holds.

These conditions are not vacuous. For any $\alpha \in (0, k/M)$, choosing $\beta > 0$ sufficiently small makes $\bar{\lambda}(\alpha, \beta, k, M) < 1$ and satisfies $(k-1)(1-\alpha) > M\beta$. Hence there exists $\lambda \in (\bar{\lambda}(\alpha, \beta, k, M), 1)$. Since the inequalities are strict, the response pattern holds on an open set of primitives.

C Simulation Appendix

C.1 Q-Learning Implementation Details

This appendix collects the exact specification of the Q-learning simulations summarized in Section 4.

Response opportunities. Formally, suppose firm i is active in period t, so firm j is scheduled next. Let ρ_j denote the probability that firm j can revise on its next scheduled turn after firm i changes price. Draw $u_{t+1} \sim U[0,1]$ and define

$$R_{j,t+1} = \mathbf{1}\{a_{i,t} \neq p_{i,t-1}\}\mathbf{1}\{u_{t+1} \leq \rho_j\}.$$

This implements the rule illustrated in Figure 7: $p_{j,t+1} = a_{j,t+1}$ and $p_{i,t+1} = p_{i,t}$ when $R_{j,t+1} = 1$, and prices remain unchanged otherwise. The main simulations vary ρ_D and ρ_S to distinguish symmetric from asymmetric response timing.

Exploration. The ϵ_t-greedy exploration rule is given in Algorithm 1, steps 2–7.

General learning target. Section 4 gives the Q-learning update rule and the simultaneous and two-turn posted-price targets. More generally, a scheduled response window may not be used, in which case the market still clears at unchanged posted prices and the firm earns another realized profit before its next pricing decision. Letting $m_i(t)$ denote the number of market clearings between firm i's choice at t and that next pricing decision, the implemented target is

$$\widehat{Q}^{rw}_{i,t}(s_{i,t}, a_{i,t}) = \sum_{k=0}^{m_i(t)-1} \beta^k \pi_i(p_{i,t+k}, p_{j,t+k}) + \beta^{m_i(t)} \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+m_i(t)}, a').$$

This reduces to the two-turn target in Section 4 when $m_i(t) = 2$ and every scheduled window is used.

Parameters. The discount factor is fixed at $\beta = 0.99$. Learning rates and exploration rates follow parallel hyperbolic decay schedules over a simulation horizon of T periods:

$$\alpha_t = \max\left\{\underline{\alpha}, \alpha_0 \frac{\kappa_\alpha}{\kappa_\alpha + t}\right\}, \qquad \epsilon_t = \max\left\{\underline{\epsilon}, \epsilon_0 \frac{\kappa_\epsilon}{\kappa_\epsilon + t}\right\}.$$

In the baseline simulations, $\alpha_0 = \epsilon_0 = 0.1$, $\underline{\alpha} = \underline{\epsilon} = 0.05$, and $\kappa_\alpha = \kappa_\epsilon = 0.1T$, so both learning and exploration decline gradually over the simulation horizon but remain bounded away from zero.

Myopic best response. Formally, the myopic best response used in the one-learner benchmarks (Section 4) is $BR_i(s_{i,t}) \in \arg\max_{p \in \mathcal{P}} \pi_i(p, p_{j,t-1})$, with ties between matching and undercutting broken toward matching, consistent with the weak inequality in equation (1).

C.2 No-Response Learning Benchmark

Figure A.4 plots the transaction-price trajectories for the simultaneous-updating benchmark discussed in the main text. With no response channel, demand asymmetry does not produce higher learned prices.

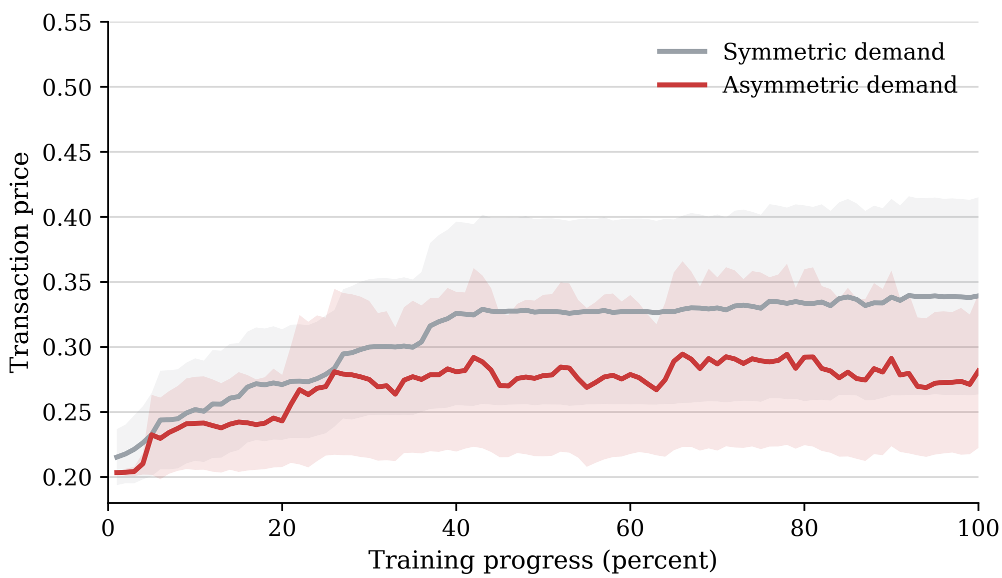

Figure A.4. Transaction Prices under Simultaneous Updating.

Notes: The figure plots average transaction prices in the simultaneous-updating benchmark. Lines average across simulation runs, and shaded regions report 95 percent confidence intervals.

C.3 Late-Period Dynamics in Two-Learner Simulations

The two-learner simulation under dominant followership generates recurrent price adjustment rather than a single fixed price pair. This is the object summarized in Table 7 and Figure 9: late-period play under mutual adaptation.

Learning nevertheless moves play toward a stable region even though the exact price pair does not settle. Figure A.5 shows that the share of periods in which both firms post prices weakly above 0.6 rises during training and is highest when the low-friction responder is also demand advantaged: play becomes concentrated in the high-price region even though the specific price pair within it continues to change.

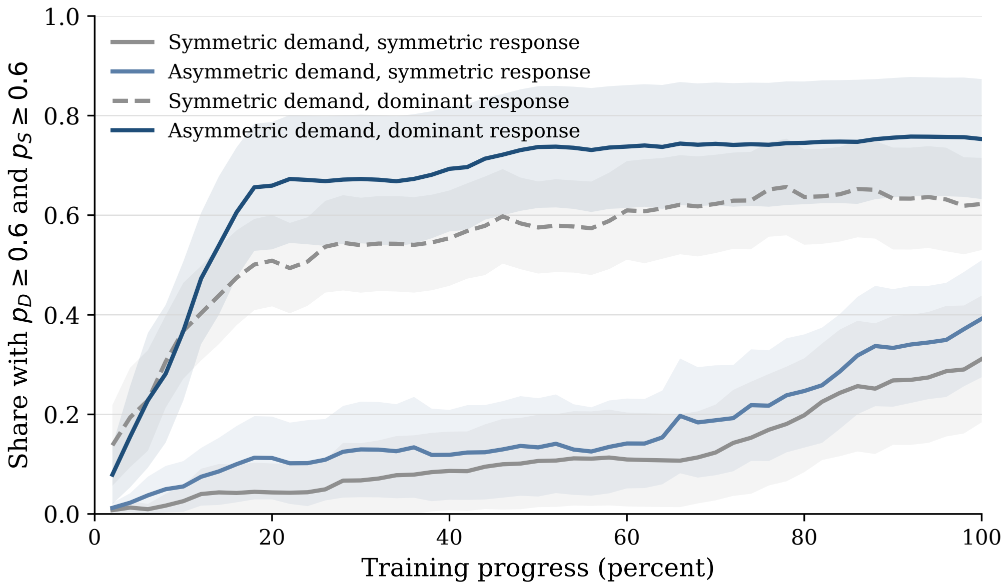

Figure A.5. Learning into High-Price States.

Notes: The figure reports, by training bin, the share of periods in which both firms post prices weakly above 0.6. Lines average across simulation runs; shaded regions report 95 percent confidence intervals across runs.

This interpretation is consistent with prior work on multi-agent Q-learning, where learned play often takes the form of recurrent price dynamics rather than a fixed price profile (Klein 2021; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024). The one-learner decomposition in Table 10 provides the cleaner evidence on the economic channel behind this pattern.

C.4 Response Decomposition

Figure A.6 decomposes firm D's next-period action after the rival posts above its inherited price. The main text focuses on matching because matching reinforces the rival's higher-price move. This figure shows what happens when matching does not occur. In the comparison cells, higher-price moves are often followed by undercutting or no response. When demand advantage and asymmetric response timing are combined, matching accounts for most of firm D's next-period actions and undercutting becomes much less frequent.

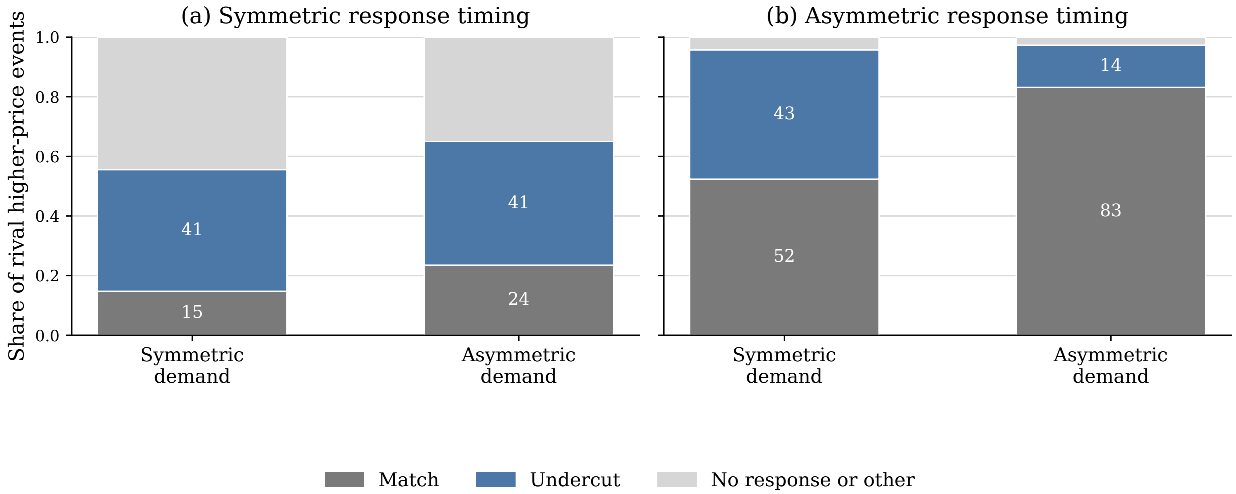

Figure A.6. Response Decomposition after Rival Higher Prices.

Notes: The figure conditions on final-5-percent periods in which firm S posts a price above firm D's inherited posted price. Bars classify firm D's next-period action. A match means firm D sets the same price as firm S; an undercut means it sets a lower price. The remaining category includes cases in which firm D's next posted price is unchanged or is above firm S's price.

C.5 Asymmetric Incidence across Runs

Figure A.7 reports the run-level counterpart to the asymmetric incidence documented in Section 4: it plots, for each simulation run, the rival's posted-price premium against the dominant firm's profit premium. Points in the upper-right region therefore correspond to runs in which the rival posts higher prices on average and the dominant firm earns higher profits. The dominant-followership runs are concentrated in this region, showing that the asymmetric incidence is systematic across runs and not an artifact of period averaging.

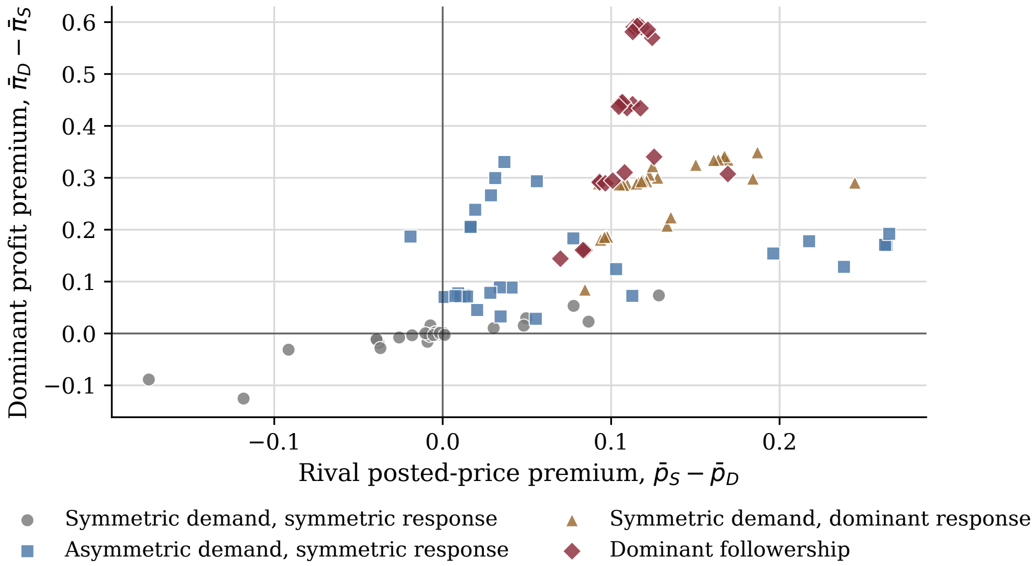

Figure A.7. Run-Level Incidence of Prices and Profits.

Notes: Each point is one simulation run, using averages over the final 5 percent of periods. The horizontal axis reports $\bar{p}_S - \bar{p}_D$, the difference between the rival's average posted price and the dominant firm's average posted price. The vertical axis reports $\bar{\pi}_D - \bar{\pi}_S$, the corresponding dominant-firm profit premium.